

University of Central Punjab

BSCS FINAL PROJECT

Requirements Specification

INTELLIGENT SECURE MADITATION COUNSELOR-PATIENT CARE SYSTEM WITH AI VIRTUAL ASSISTANT



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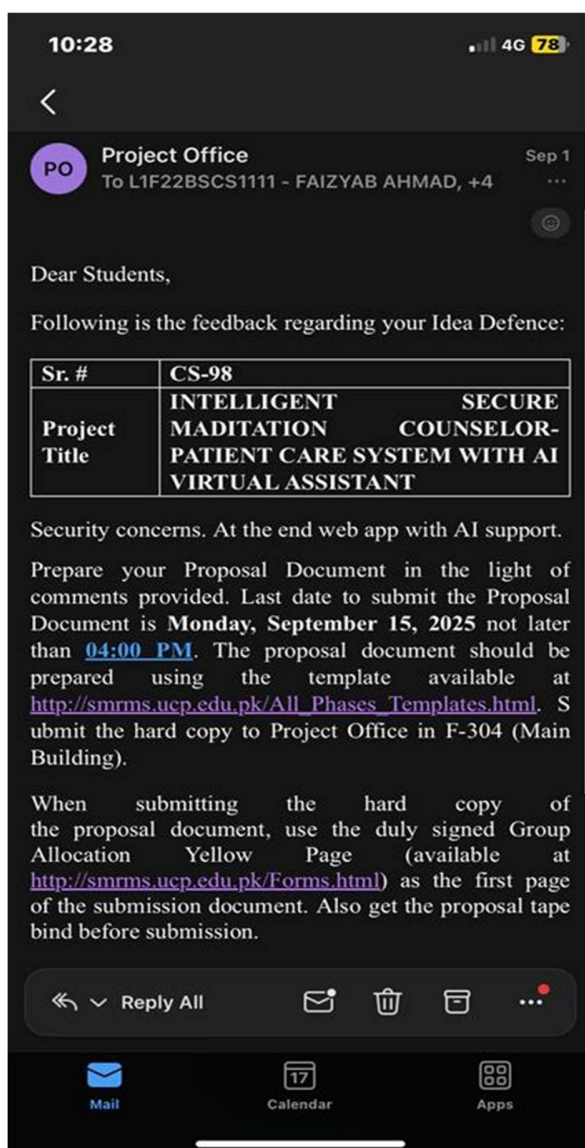
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Revision History

Name	Date	Reason For Changes	Version

Previous Phases Feedback

Idea Defense Feedback (Screenshot)



Abstract

The design and development of a reliable, artificial intelligence-powered online platform aimed at enhancing and speeds up the operation of therapeutic relationships and patient care is discussed in this project. To meet the needs of all major stakeholders, the system is designed around three different, role- based interfaces. With the help of artificial intelligence insights, the platform gives therapists; the ability to manage patient appointment schedules, medical records, and session notes. The interface makes it easier for patients to find and schedule appointments with therapists, take part in encrypted video sessions from beginning to end, monitor therapeutic tasks and personal health metrics, and communicate with a 24/7 personalised chatbot assistant. A specialised portal for professional account management combines administrative tasks. The system's context-aware AI virtual assistant is one of its main innovations. This module creates customised useful recommendations for patients and therapists by analysing longitudinal patient data, such as medical history, allergies, clinical notes, and personal preferences. Additional features that enhance engagement and positive outcomes include gamified goal tracking with a points system, automated reminders, integrated payment processing, an anonymous interaction mode, and visual progress analytics. Secure encryption, strict role-based access control (RBAC), and secure authentication protocols are all part of the platform's foundational security and privacy development. This complete security framework promotes a trusted environment for the delivery of digital healthcare by guaranteeing the security and privacy of sensitive health information.

1. Introduction and Background

The web-based, extremely secure Intelligent Therapist-Patient Management System with AI Virtual Assistant was created especially to improve communication between patients and therapists, streamline the administration of therapy sessions, and guarantee safe communication. Effective therapy management continues to be a major challenge as the importance of mental health care grows. Many therapists still rely on labor-intensive, time-consuming manual processes like phone calls and documentation on paper, which are prone to scheduling errors. In the meantime, patients frequently struggle to schedule sessions, monitor their progress, and protect their privacy while receiving therapy. By offering a centralized, intelligent system that allows therapists to register patients, manage extensive medical records, create flexible schedules, and conduct encrypted online sessions, this project tackles these issues. To reduce judgement and motivate more people to seek help, patients can quickly locate therapists, check real-time availability, schedule appointments, pay, and stay anonymous during sessions. One notable feature is the AI Virtual Assistant, which provides ongoing support outside of sessions by sending tailored recommendations and reminders based on each patient's unique profile, which includes medical history, allergies, and therapist notes. The system's integration of automation, artificial intelligence-powered personalization, and strong privacy protections not only reduces the administrative burden on therapists but also enhances patient engagement, therapy plan adherence, and overall therapy results. This innovative platform creates a more effective, efficient, and reliable therapy setting by satisfying the growing need for digital, easily accessible, and private mental healthcare solutions. The complete software requirements that follow in this document are firmly established by this introduction.

1.1 Product (Problem Statement)

Effective therapy delivery and patient engagement are complicated by several serious problems facing the leadership of mental health care today. Therapists frequently use manual procedures like phone-based scheduling and paper records, which leads to inefficiencies, time consumption, and frequent scheduling errors. Patients may be avoided from seeking or continuing treatment if they have trouble scheduling appointments, monitoring their therapeutic progress, and maintaining privacy during delicate sessions. Current digital platforms frequently don't offer strong privacy features like anonymous communication options or individualized support based on each user's medical history. These problems result in missed appointments, reduced therapy adherence, therapist burnout, and generally less than ideal mental health outcomes. A secure, automated, and intelligent web-based platform that enables smooth therapist-patient interactions, real-time scheduling, encrypted communication, and AI-driven personalized support is what the Intelligent Therapist-Patient Management System seeks to address. This system aims to increase the accessibility, effectiveness, and reliability of mental

health therapy by eliminating inefficiencies, boosting privacy, and boosting engagement.

1.2 Background

This project focusses on modern digital mental healthcare, a developing field that addresses significant inefficiencies in conventional therapy by combining telehealth and medication management. Although remote sessions have made existing platforms more accessible, they frequently function in silos and provide separate solutions with little to no integration between a patient's continuing care journey and a therapist's clinical tools. This gap leads to impersonal patient care, stressful administrative workloads for professionals, and enduring privacy issues that discourage people from getting treatment. Within this framework, our project seeks to go beyond traditional teletherapy by developing a single, safe ecosystem that intelligently automates processes, encourages team progress monitoring, and provides genuinely customized, context-aware AI assistance to improve patient engagement and therapeutic results.

1.3 Scope

The creation of an extensive, web-based platform intended to monitor and protect the entire therapist- patient relationship is covered by this Software Requirements Specification (SRS). Among the features included are a Therapist Portal for scheduling, secure storage of medical records, and patient registration; a Patient Portal for viewing records, scheduling appointments in real time with integrated payments, and tracking progress through a gamified goal system; and a core AI Virtual Assistant that analyses patient profiles to provide context-aware, customized recommendations. Additionally, a mutual feedback mechanism, an anonymous interaction mode for improved patient privacy, and secure, end-to- end encrypted video consultations will all be made possible by the system. Providing independent AI-led therapy, creating native mobile applications (although the web app will be responsive), integrating hardware, performing direct diagnostics, and managing insurance claim processing are all specifically outside the purview of this project.

1.4 Objective(s)/Aim(s)/Target(s)

This project's main goal is to create a **Secure Meditation Counsellor Patient Care System with AI** integration that transforms mental health consultations by offering intelligent automation, individualized virtual counselling, and robust privacy protection. Therapists can use the system to assign wellness goals, schedule and carry out encrypted therapy sessions, securely manage patient records, and track patient progress. In addition, it gives patients access to real-time therapy updates and recommendations, anonymous interaction options, and personalized AI-driven support, all from a single, secure web platform.

1.4.1 Develop a Secure Therapist–Patient Management Platform

To create a centralized portal with encrypted storage and restricted access that enables therapists to register, manage, and keep an eye on patient profiles and therapy sessions while ensuring total data confidentiality.

1.4.2 Implement Automated Appointment Scheduling

To reduce scheduling conflicts and no-shows, a dynamic booking system that automatically verifies therapist availability, avoids double booking, and provides real-time confirmations should be designed.

1.4.3 Integrate Secure Communication and Consultation Channels

To provide an Anonymous Interaction Mode for encrypted video and text-based consultations, protecting patients' privacy who would rather remain anonymous during treatment sessions.

1.4.4 Incorporate an AI-Powered Virtual Assistant

To incorporate an intelligent assistant that can evaluate session data, therapist notes, and patient medical histories to offer tailored, context-aware advice and reminders about mental health.

1.4.5 Enable Goal Assignment and Progress Tracking

To enable patients to track their progress using a gamified points-based system that promotes engagement, and to let therapists assign customizable wellness or therapeutic goals.

1.4.6 Establish Secure Medical Record Storage

To keep a secure database of therapy notes, prescriptions, and clinical reports that is only accessible by authorized personnel and complies with accepted data security procedures.

1.4.7 Implement Robust Authentication and Access Control

To create a secure authentication system that protects user data from unwanted access by utilizing session management, role-based access control (RBAC), and encrypted credentials.

1.4.8 Ensure Comprehensive System Security

Enforcing secure API communication, end-to-end encryption for data in transit and at rest, and ongoing monitoring to preserve information integrity, confidentiality, and dependability.

1.4.9 Automate Notifications and Reminders

To enhance communication by using integrated email and SMS services to automatically notify people about therapy exercises, upcoming appointments, and medication schedules.

1.4.10 Provide Data-Driven Progress Analytics

To produce AI-based reports and visual progress dashboards that let patients and therapists assess progress, spot trends in behavior, and modify treatment plans as necessary.

1.4.11 Enhance Accessibility and User Experience

To create a user-friendly, responsive, and intuitive web interface that is optimized for a variety of devices, guaranteeing smooth experience for patients and therapists of all technical backgrounds.

1.5 Challenges

To guarantee system effectiveness, dependability, and user trust, several real-world issues must be resolved in the development of the AI-Integrated Secure Meditation Counsellor Patient Care System. The primary challenges that are expected to arise are:

1.5.1 Ensuring Data Privacy and Security

One of the main concerns is protecting private information between patients and therapists. It can be difficult to implement stringent access control, secure authentication, and end-to-end encryption while preserving seamless operation.

1.5.2 Balancing AI Assistance with Human Therapy

Designing an AI assistant that supports rather than replaces therapists is a key challenge. The system must offer tailored advice without making direct medical diagnoses or transgressing moral boundaries.

1.5.3 Reliable Real-Time Communication

It can be challenging to keep therapist-patient video or chat sessions steady and secure, particularly when there is a high system load or a limited network.

1.5.4 Implementation of Anonymous Mode

It is technically and procedurally difficult to create an anonymous mode that preserves patient identity while guaranteeing accountability and correct record keeping.

1.5.5 User Trust and Adoption

Strong guarantees of data security, user-friendliness, and steady system dependability are necessary to inspire patients and therapists to embrace AI-supported counselling systems with confidence.

1.6 Learning Outcomes

The creation of the AI-Integrated Secure Meditation Counsellor–Patient Care System has yielded significant professional and technical learning opportunities. Working with technologies like React.js, Node.js, Express.js, and MongoDB [6] allowed for the development of a dynamic and secure web platform while also providing a thorough understanding of full-stack web development. The use of data encryption, authorization, and authentication methods improved knowledge of system security and privacy protection in delicate medical applications. Including an AI-powered virtual assistant provided hands-on experience with AI concepts such as contextual response generation and data-driven personalization. Furthermore, working with

real-time communication systems enhanced comprehension of socket-based and API-driven video and chat integration for real-time interactions between patients and therapists. Additionally, the project improved skills in project management and system design, including deployment procedures, version control, and modular architecture design. Beyond technical proficiency, the experience deepened understanding of ethical issues in AI-based mental health systems, especially those pertaining to data security, confidentiality, and responsible AI use. Overall, this project showed how technology can be used responsibly to improve the delivery of mental health care while also greatly enhancing analytical, problem-solving, and development skills.

1.7 Nature of End Product

An intelligent and safe platform for managing therapist-patient interactions in the realm of mental health and wellness will be the project's finished product, a fully functional web-based application. To protect privacy and data, it will have distinct interfaces for patients and therapists with role-based access. Patients will have access to their own dashboards to view therapy details, track assigned goals, and securely communicate with their therapists, while therapists will be able to register and manage patient profiles, schedule appointments, store medical reports, and assign wellness tasks. The system will incorporate artificial intelligence (AI)-powered virtual help that can offer customized recommendations and prompts based on each patient's unique profile and treatment status. To establish a smooth digital therapy environment, essential features like anonymous interaction, real-time chat and video consultation, and secure payment and notification systems will be integrated. The finished product will be scalable, responsive, and easy to use on a variety of devices. The final product will essentially function as a complete, privacy-focused, and intelligent counselor-patient management solution that combines data security, automation, and individualized AI-driven care with mental health support.

1.8 Completeness Criteria

When all intended features and functional elements of the AI-Integrated Secure Meditation Counselor Patient Care System have been fully developed, tested, and put into use in accordance with the system requirements, the project will be deemed finished. Therapists must be able to safely register patients, schedule and carry out therapy sessions via encrypted video or chat, assign wellness goals, and manage medical records using the system. It should be possible for patients to track their progress, view therapy notes, log in with authenticated credentials, access their dashboards, and securely communicate with therapists, including through the anonymous interaction mode. Based on stored data, the AI virtual assistant must accurately provide context-aware, personalized reminders and suggestions. Every important module should function dependably and without significant errors, including notifications, AI integration, goal tracking, appointment scheduling, encryption, and authentication. The user

interface needs to be accessible on all devices, completely responsive, and easy to use. To guarantee system stability and data protection, successful completion will also necessitate passing functional, performance, and security testing. The project will be considered finished once all these goals have been met and verified by evaluation and demonstration.

1.9 Business Goals

The AI-Integrated Secure Meditation Counselor–Patient Care System's main objective is to develop a long-lasting digital platform that enables therapists to better manage their practices and reach a wider clientele. By automating crucial processes like patient registration, appointment scheduling, session management, and progress tracking, the system seeks to improve the efficiency of mental health practitioners' workflow. The platform's subscription-based business model will bring in recurring income, and free trial periods will be offered so that therapists can test out its features and usability before deciding on a plan. Additionally, the platform acts as a discovery tool, assisting patients in finding and contacting licensed therapists, which raises the profile and commercial prospects of mental health professionals. Strategically speaking, the system is built for long-term scalability, with the potential to develop into a mobile application and accommodate ongoing updates for improved functionality and user satisfaction. The overall objective of the business is to create a dependable, safe, and intelligent ecosystem that facilitates the expansion of therapists' practices and gives patients convenient access to expert mental health care.

1.10 Related Work/ Literature Survey/ Literature Review

Several online platforms for mental health and therapy have surfaced in recent years to help patients connect with licensed therapists. However, the majority of these platforms, which frequently lack a cohesive system that combines security, automation, and personalisation through artificial intelligence, concentrate primarily on either practice management or patient–therapist communication. By integrating crucial features like anonymous treatment, AI-driven guidance, secure communication, and automated therapy management into a single, all-inclusive solution, my project, the AI-Integrated Secure Meditation Counselor–Patient Care System, seeks to overcome these constraints.

Through online chat and video sessions, users can connect with licensed therapists through existing platforms such as BetterHelp and Talkspace. Although they offer a scalable and convenient setting for mental health consultations, patients, not therapists, typically pay a subscription fee. These platforms have also come under fire for their lack of individualized AI support and data privacy. Systems like TheraNest, on the other hand, are more practice management-oriented and provide tools for basic telehealth, scheduling, billing, and documentation. These systems work well for administrative purposes, but they don't give patients access to AI-based engagement or anonymous interaction options. Furthermore,

through conversational interfaces, AI mental health chatbots like Woebot and related applications show how AI can promote mental health. These tools, however, are not associated with licensed therapists and are unable to provide individualized data-based recommendations or clinical treatment.

A comparison of the current systems and how my suggested solution overcomes their drawbacks is shown in the following table:

Feature	BetterHelp Talkspace	Woebot (AI Chatbot)	TheraNest (Practice Management)	Proposed System
Therapist Discovery	✓	✗	✗	✓
AI Virtual Assistant	✗	✓	✗	✓
Anonymous Mode	✗	✓	✗	✓
Secure Data Storage	✓	✓	✓	✓ (Enhanced Encryption)
Real-Time Chat / Video	✓	✗	✓	✓
Goal Tracking	✗	✓ (Limited)	✗	✓
Therapist Subscription Model	✗	✗	✗	✓
Personalized Suggestions	✗	✓ (Generic)	✗	✓ (Context- Aware)

Table 1: Comparison of Existing and Proposed Systems

In conclusion, most current platforms prioritize either therapist management or patient convenience, but not both at the same time. Additionally, they don't incorporate an AI assistant that can make recommendations based on user profiles that are both context-aware and privacy-preserving. By integrating AI-powered personalized support, therapist workflow automation, and secure patient data management into a single, scalable platform, my suggested system closes this gap. Additionally, it is technically innovative and commercially

viable product due to its free trial offering, mobile scalability, and subscription-based model for therapists. Patients can confidently access therapy in a secure, private, and encouraging digital environment with this approach, which also guarantees that therapists can effectively manage their practices while reaching more clients.

1.11 Document Conventions

The IEEE Software Requirements Specification (IEEE 830-1998) guidelines served as the model for the structured and standardized format used in this Software Requirements Specification (SRS) document. For readability, accuracy, and convenience of reference, it keeps its layout simple and uniform. Throughout the document, the following conventions have been adopted:

- **Font Style and Size:**
The document is written using *Arial And Times* font, with headings in **bold (size 14)** and main text and sub heading in **regular (size 12)** for professional presentation and clarity.
- **Numbering Scheme:**
All sections and subsections are numbered hierarchically (e.g., 1.1, 1.2, 1.3, etc.) to provide clear organization and logical flow of information.
- **Terminology:**
Key terms such as *Therapist*, *Patient*, *System*, *AI Assistant*, and *Anonymous Mode* are capitalized to indicate system entities or core components within the project.
- **Symbols and Notations:**
Standard symbols ($\checkmark$, X) are used in comparison tables to indicate the presence or absence of specific features. Acronyms such as **AI** (Artificial Intelligence), **API** (Application Programming Interface), and **RBAC** (Role-Based Access Control) are expanded when first introduced.
- **Language Style:**
The document uses formal, technical, and concise language suitable for academic and professional contexts. Passive voice is used where appropriate to emphasize process and functionality.
- **Figures and Tables:**
Figures and tables are labelled sequentially (e.g., *Figure 1*, *Table 1*) and referenced within the text where relevant to maintain coherence and traceability.

- **Emphasis:**
Important terms and definitions are either bolded or italicized to draw attention to key points or technical terminology.

These conventions ensure that the document maintains a consistent format, allowing readers, developers, and evaluators to interpret the content accurately and efficiently.

Overall Description

2.1 Product Features

The AI-Integrated Secure Meditation Counselor–Patient Care System is an online platform that uses artificial intelligence, automation, and security to improve the therapist-patient relationship. It offers a single setting for managing therapy, communicating in real time, and interacting with patients. Below is a summary of the main features of the product:

1.10.1 Secure User Authentication and Role-Based Access:

Both patients and therapists can safely log in to the system. Therapists can manage patient records while patients can only view their own data thanks to access rights that are defined according to user roles. To ensure security and privacy, all data transactions are encrypted.

1.10.2 Therapist Dashboard:

Therapists can register new patients, manage existing profiles, set up appointments, take session notes, upload medical reports, and track progress using a dedicated dashboard. This increases overall workflow efficiency and decreases manual paperwork.

1.10.3 Patient Portal:

To view assigned therapy goals, scheduled sessions, and therapist feedback, patients can access their personal dashboard by logging in. Additionally, they can visually monitor their progress over time and securely communicate via integrated chat or video sessions.

1.10.4 Appointment Scheduling and Management:

An automated appointment scheduling module that shows the availability of therapists in real time is provided by the system. It has integrated payment processing for session fees, avoids duplicate bookings, and sends automated reminders.

1.10.5 AI Virtual Assistant:

Patients can receive context-aware, personalized recommendations from an intelligent AI assistant. It can provide health-related insights based on medical history and therapist input, remind users of therapy goals, and recommend relaxation techniques or meditation.

1.10.6 Anonymous Interaction Mode:

The system has an Anonymous Mode that lets users participate in video or chat therapy sessions without disclosing who they are, protecting patient privacy and lowering stigma.

1.10.7 Secure Data Management:

Encryption techniques are used to safely store all sensitive data, including reports,

prescriptions, and therapy notes. Strict access controls guarantee adherence to confidentiality and privacy laws.

1.10.8 Notifications and Reminders:

In order to keep patients and therapists organized and consistent, automated email or SMS notifications are sent for tasks, therapy milestones, and upcoming appointments.

1.10.9 Progress Tracking and Analytics:

Patients can track advancements in their therapeutic journey with the aid of the system's visual progress reports. With AI-based insights, therapists can examine patterns in patients' progress, allowing for better decision-making and treatment modifications.

1.10.10 Scalable and Responsive Design:

The platform's responsive and scalable design guarantees device compatibility and permits future growth into mobile applications for increased accessibility.

2.2 User Classes and Characteristics

Three primary user classes: Admin, Therapist (Doctor), and Patient will utilize the system. Every class has different roles, levels of access, and ways of interacting with the system. To make the platform easy to use for both technical and non-technical users, the overall design places a strong emphasis on accessibility, simplicity, and secure data handling.

2.2.1 Admin

Description:

The supervisor is in the position of monitoring system-level functions, specifically managing user access control, subscriptions, and therapist account verification. This function makes sure that the platform's services are only available to patients and therapists who are authorized.

2.2.2 Key Characteristics:

Offers complete access to features for managing subscriptions (adding, removing, starting, or ending plans).

- To protect patient privacy and data security, no sensitive patient information may be viewed or accessed.
- Manages and maintains an eye on payment verification, user onboarding, and system health.
- It requires a moderate level of technical expertise and is used occasionally rather than daily, mostly for administrative updates and subscription management.

2.2.3 Therapist (Doctor):

The system's primary service providers are therapists. The platform is used for practice

management, patient registration, scheduling, therapeutic task assignment, patient progress review, and secure video sessions.

2.2.4 Key Characteristics:

Makes frequent use of the system (daily or several times a day).

2.3 Engages with dashboards that display reminders, appointments, and patient updates.

2.4 Able to assign customized tasks and track real-time patient progress.

2.5 Capable of safely storing and reviewing reports, prescriptions, and session notes.

2.6 Possess integrated communication tools, including encrypted video sessions and chat.

2.7 Needs an intuitive, easy-to-use interface that reduces navigational effort.

2.8 The interface is meant to be self-explanatory and intuitive, so a moderate level of technical expertise is required.

2.2.5 Importance Level:

Very High The system's main clients are therapists, and their contentment directly affects the system's overall performance.

2.2.6 Patient Description:

In addition to their therapist, patients are the end users who interact with the system and receive therapy. Simple, entertaining, and appropriate for users of all ages including elderly or non-technical patients, is how the interface was created.

2.2.7 Key Characteristics:

- Has the ability to view assigned tasks, progress, reminders, and session schedules on personal dashboards.
- Uses a user-friendly interface with emojis or icons to interact with the AI virtual assistant.
- During therapy sessions, Anonymous Mode can be enabled to safeguard privacy and identity.
- Uses game-like interactivity, to-do lists, reminders, and progress tracking to keep users motivated.
- Usually make use of the system once a week or according to the session schedule.
- Needs little technical knowledge; the user interface prioritizes ease of use and clear visibility of key options (tasks, chatbot, progress).

2.2.8 Importance Level:

High Reliability of the platform and successful therapy engagement are guaranteed by patient satisfaction.

2.2.9 Favored User Classes:

All user classes are necessary, but since they are the platform's main customers, therapists are the most important to satisfy. To maintain engagement and trust, it is equally important to give patients a simple and inspiring experience. Sometimes being less involved, the manager's position guarantees system security, subscription management, and operational continuity.

2.3 Operating Environment

High availability, cross-platform compatibility, and the smooth integration of AI-driven automation are all guaranteed by the secure, cloud-based web environment in which the Intelligent Secure Meditation Counsellor-Patient Care System with AI Virtual Assistant will function. Both technical and non- technical users can access the system because it is made to function well across a variety of hardware, operating systems, and browsers.

2.3.1 Hardware Platform:

- The application will be set up on a cloud-based hosting server that supports Python, MongoDB, and Node.js.
- Standard computing devices like desktops, laptops, tablets, or smartphones will be used by end users (the administrator, therapist, and patient) to access the platform.

2.3.2 Suggested minimal requirements for end users:

- Processor: Dual-core 1.8 GHz or faster
- RAM: 4 GB or more
- Stable broadband or 4G/5G connectivity for video sessions and real-time communication.

2.3.3 Operating Systems:

- Server-Side: Works with Windows Server 2019 and later or Linux-based servers (such as Ubuntu 20.04 LTS or higher).
- Client-Side: Compatible with most major operating systems, such as:
 - Windows 10 / 11
 - macOS Monterey or later
 - Android 10 or later
 - iOS 15 or later

2.3.4 Software Components:

Frontend: React.js, HTML5, CSS3, JavaScript, and, if desired, Tailwind CSS or Bootstrap 5 for responsive design were used in its development.

Backend: Node.js and Express.js for database communication, authentication, and API

endpoint management.

Database: MongoDB for safe and effective user data, health record, and progress log storage. Personalized recommendations, chatbot responses, and task suggestions based on patient profiles are handled by the AI Processing (Python Module), which is constructed using Flask or Fast API.

Automation and Workflow Integration: n8n [10], a workflow automation tool that links Python, Node.js, and external APIs, was used to implement this. n8n will: At predetermined intervals (daily or weekly, for example), initiate AI-driven recommendations. At scheduled times (daily or weekly, for example), initiate AI-driven recommendations. Automate progress reports and reminders. Connect to external APIs such as calendar services, email, SMS, and Zoom [8]. Use REST API calls to safely send and receive data between the Python AI module and the Node.js backend.

Communication Services: Google/Zoom Integration [18] Use Socket.IO for chat and real-time notifications and meet for secure video consultations.

Version Control: GitHub and Git are used to track deployment and facilitate collaborative development.

2.3.5 Browser Compatibility:

All the main browsers will be able to use the web application, including:

- Google Chrome (version 90 or higher)
- Mozilla Firefox (version 85 or higher)
- Microsoft Edge (version 90 or higher)
- Apple Safari (version 14 or higher)

2.3.6 Security and Environment Constraints:

- HTTPS and SSL/TLS encryption will be used to secure all communications between the client, backend, and n8n workflows.
- Data isolation between user roles (Admin, Therapist, Patient) will be guaranteed by access controls.
- Python 3.9 or later, Node.js v18 or later, MongoDB v6 or later, and n8n for automation orchestration will all be supported by the hosting server.
- Regular backup procedures and ongoing integration will preserve system uptime and reliability.

2.4 Design and Implementation Constraints

Several design and implementation limitations will affect the creation and use of the Intelligent Secure Meditation Counsellor-Patient Care System with AI Virtual Assistant. These limitations guarantee that the system will continue to be safe, scalable, privacy-compliant, and able to operate dependably in the designated technical environment. The system will be implemented on a cloud-based platform, which provides scalability and flexibility but also necessitates compliance with network latency considerations, server resource limitations, and API configurations. Python, MongoDB, Node.js, and n8n workflows must all be supported simultaneously in the cloud environment. The hosting server must have reliable uptime and support asynchronous communication protocols like WebSocket and REST APIs because several services like AI processing, scheduling automation, and encrypted video sessions will interact in real time. Compliance with privacy and data security regulations is a significant design limitation. All data storage, processing, and communication must adhere to international data protection standards like HIPAA [4] and GDPR [2], since the system handles sensitive patient-therapist data. Personal data must be anonymized, access controlled and encrypted from beginning to end. To maintain confidentiality and compliance with legal and ethical standards for handling mental health information, the administrator role is purposefully prohibited from accessing any patient-related data. Instead of operating as a stand-alone service, the Python-developed AI module will be integrated through n8n workflows. This method makes scheduling and automation easier, but it also adds dependencies on the timing of workflow execution and the stability of the n8n server. The integration will use RESTful communication, with data sent securely over HTTPS, between the Python model endpoints and the Node.js backend. Another limitation is imposed by external dependencies. For scheduling and communication, the system uses third-party APIs like Zoom, Google Meet, email, and SMS services. System reliability may be impacted by their rate limits, availability, and authentication procedures. To ensure that scaling requirements can be satisfied without going over usage limits, these limitations will be lessened by implementing suitable subscription tiers for the APIs. The core development team will maintain the project during early deployment, while a specialised technical team working under the direction of the original developers will handle long-term maintenance following scaling. To guarantee a seamless transition and maintainability, this calls for strict adherence to coding standards, version control via Git/GitHub, and unambiguous documentation. In conclusion, the design needs to strike a balance between stringent data privacy regulations, third-party API dependencies, cloud performance limitations, and the integration overhead of automation tools like n8n. Together, these restrictions establish the technical parameters that will govern the system's development, deployment, and upkeep to guarantee safe, reliable, and expandable operation.

2.5 Assumptions and Dependencies

Several expectations and outside dependencies are necessary for the Intelligent Secure Meditation Counsellor-Patient Care System with AI Virtual Assistant to be implemented and run effectively. Any departure from these presumptions could influence the system's security, dependability, or performance. They are based on the current design plan and available

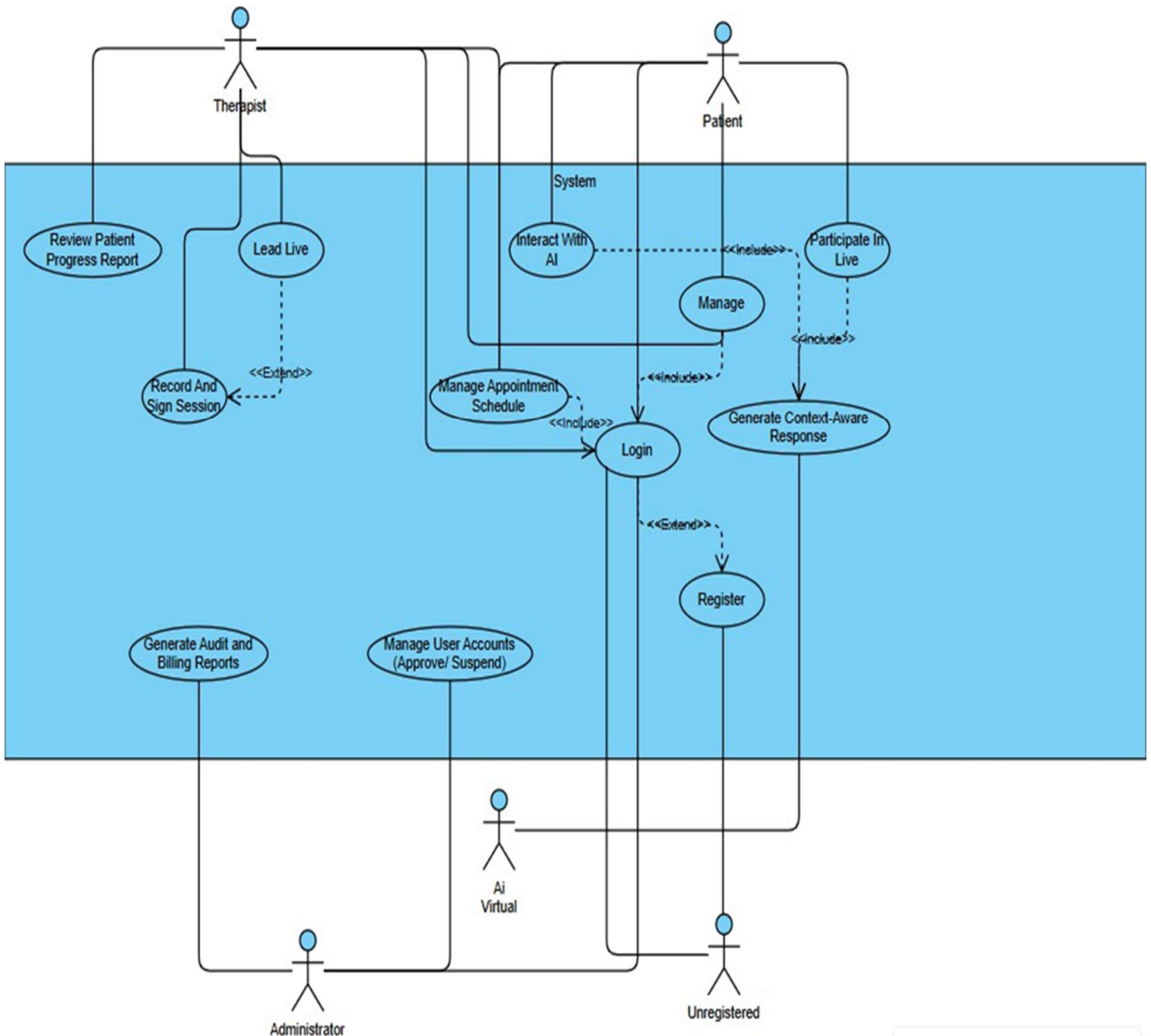
technologies. It is assumed that the cloud hosting environment will have enough processing power to support the Python AI module, Node.js backend, MongoDB database, and workflow automation all at once. For real-time functions like video conferences, live chat, and automated AI recommendations, a steady network connection and minimal latency are necessary. The system anticipates minimal service interruptions and constant uptime from the hosting provider. Another important presumption is that third-party APIs, such as Zoom, Google Meet, email, and SMS services, will continue to be reliable, well-documented, and reachable via safe authentication techniques. The project's essential scheduling and communication features rely on these services. To keep the system compatible, it might be necessary to make changes or upgrade subscription tiers if any of these APIs alter their pricing, access policies, or data exchange formats. Additionally, the dependability of n8n automation workflows and the platform's compatibility with Python and Node.js environments are assumed. These processes are essential for automated notifications, reminders, and AI-driven tasks that are scheduled. Automation performance may be momentarily impacted by any significant updates or outages in n8n until they are fixed. The project assumes that users will have access to up-to-date web browsers and devices that can support interactive dashboards and encrypted video calls. Current iterations of Chrome, Firefox, Edge, and Safari are all compatible with the front-end design. The features of the system might not be fully supported by outdated browsers. Regarding data security, it is presumed that the hosting server will consistently maintain SSL/TLS certificates and secure HTTPS connections. For the project to comply with privacy regulations like HIPAA or GDPR, encryption standards and access control measures must be consistently implemented. From a development standpoint, the system relies on the dependability of open-source frameworks like Python Flask or Fast API, React.js, Node.js, Express.js, MongoDB, and their corresponding libraries. System maintenance may be impacted, and refactoring may be necessary if these technologies undergo significant updates, deprecations, or incompatibilities. Lastly, effective teamwork and version control are essential to the project's long-term viability. To ensure smooth scaling and handover, it is assumed that all future developers or maintainers will adhere to the same documentation practices, coding conventions, and Git or GitHub workflows set up by the original team. In conclusion, the project's expansion and functionality rely on the ongoing dependability of open-source technologies,

automation tools, third-party APIs, cloud infrastructure, and adherence to secure communication standards. To maintain system integrity and sustain performance, any major change in these dependencies may necessitate matching operational or technical changes.

3 Functional Requirements

All functional requirements are expressed as use cases. The system's core functionality revolves around managing therapist-patient interactions securely, efficiently, and intelligently through automation and AI support. Each use case identifies a specific task performed by system actors (Admin, Therapist, Patient, or AI Assistant).

Figure: Primary Use Case Diagram



3.1 Book Appointment with Therapist

Identifier	UC-1	
Purpose	To allow a patient to view therapist profiles, check their real-time availability, and securely book an appointment through integrated payment and automated confirmation.	
Priority	High	
Pre-conditions	1. Patient must be registered and logged into the system. 2. Therapist profile must be verified and available in the system. 3. Stable internet connection must be active.	
Post-conditions	1. Appointment details are stored in both patient and therapist dashboards. 2. Automatic confirmation message and reminders are sent via email/SMS.	
	3. The therapist's schedule is updated to prevent double bookings.	
Typical Course of Action		
S#	Actor Action	System Response
1	Patient logs into the portal and opens the "Find Therapist" section.	System displays all verified therapist profiles along with specialization, ratings, session fee, and available slots.
2	Patient selects a therapist and clicks "Book Appointment."	System loads the therapist's real-time calendar showing available slots.
3	Patient chooses the desired date and time for the session.	System validates the selected slot and checks for conflicts.
4	Patients choose a payment method (Online or Pay in Person) and proceeds to confirmation.	System either redirects to a secure payment gateway (for online) or marks the slot as "Partially Confirmed" (for pay in person).
5	Patient confirms the booking.	System updates both patient and therapist dashboards, sends confirmation notifications, and schedules automated reminders.
Alternate Course of Action		
S#	Actor Action	System Response
1	Patient selects a time slot that has already been taken.	System displays "Slot Unavailable" and prompts the

		user to choose another time.
2	Payment fails or is cancelled mid-process.	System shows "Payment Failed – Please Try Again" and logs the event for audit purposes.
3	Therapist cancels or reschedules the session later.	System automatically notifies the patient, offers rescheduling options, and updates both dashboards
4	Patient selects "Pay in Person" instead of online payment.	System reserves the slot and marks it as "Partially Confirmed" , notifying both patient and therapist that payment will be completed at the clinic before the session.

This case describes how a patient schedules a therapy session using the system. The patient can look through therapist profiles after logging in, which contain all the information they need, including availability, reviews, fees, and expertise. After selecting a qualified therapist, the system confirms available timeslots and offers two ways to make a reservation: Pay in Person (for partial confirmation) or Online Payment (for instant confirmation). The chosen time slot is reserved by the system and marked as "Partially Confirmed" until payment is made in person if the patient chooses to pay in person. Both parties receive automated notifications upon successful booking or payment, and the therapist's calendar is updated instantly. Through automated alerts and rescheduling options, the system gracefully handles exceptions such as failed transactions, double bookings, or therapist cancellations. For both patients and therapists, this procedure maintains openness and dependability while guaranteeing efficient, adaptable, and safe appointment scheduling.

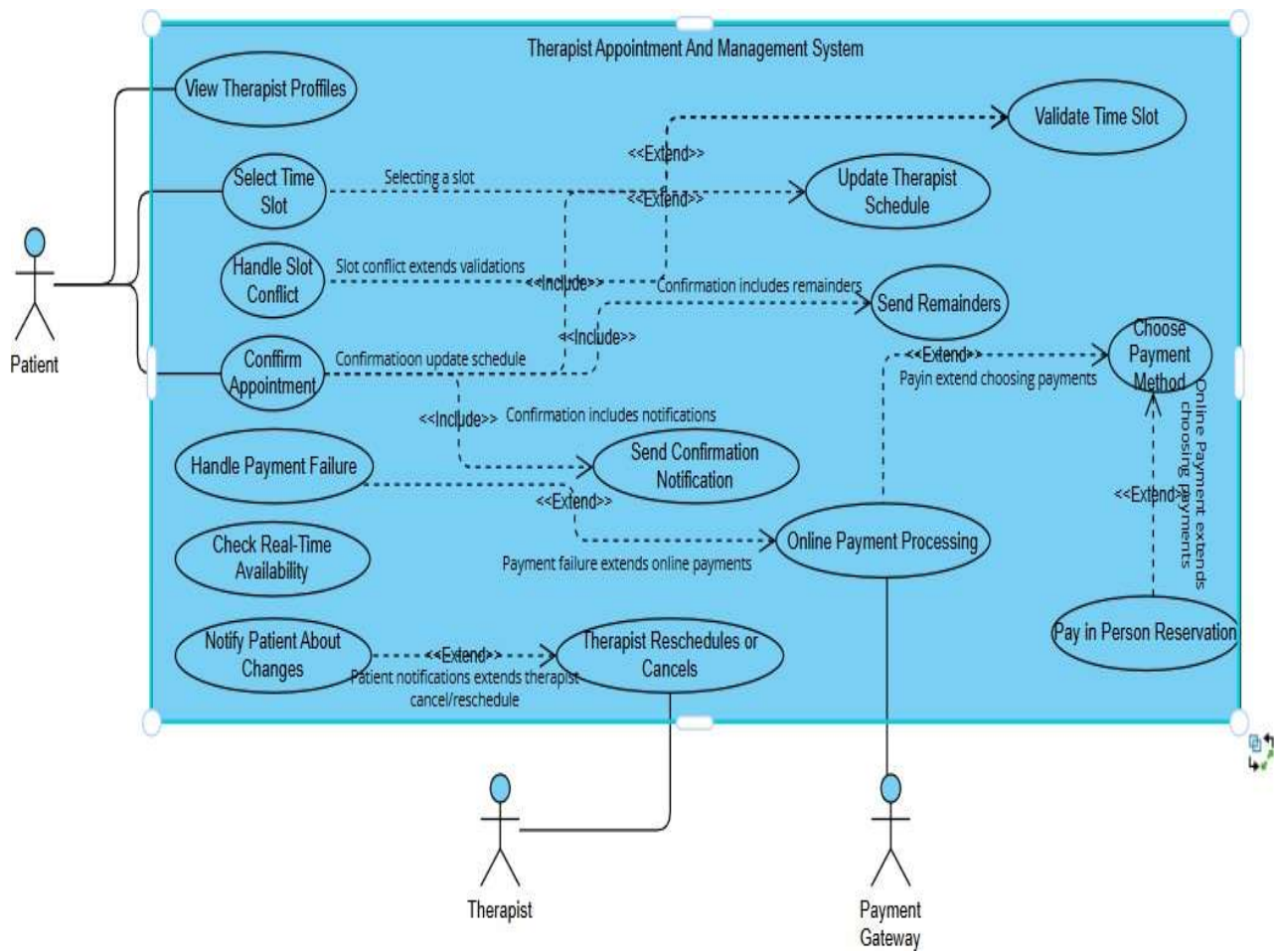


Figure: Use Case 1

3.2 Conduct Therapy Session (Encrypted + AI Support)

Purpose	To enable therapists and patients to conduct secure video or chat sessions with encryption and optional AI assistance for recommendations during or after sessions.	
Priority	High	
Pre-conditions	Both patient and therapist must be logged in. Appointment must be confirmed. Video session tools (Zoom/Google Meet) integrated.	
Post-conditions	Session summary recorded. Therapist notes and AI insights stored in database. Notifications sent post-session.	
Purpose	To enable therapists and patients to conduct secure video or chat sessions with encryption and optional AI assistance for recommendations during or after sessions.	
Typical Course of Action		
S#	Actor Action	System Response
S#	Actor Action	System Response
1	Patient and therapist log in and navigate to session dashboard.	System displays "Join Session" button for both.
2	Therapist starts session.	System opens encrypted video/chat room.
3	AI assistant monitors session context (if enabled).	System records key points and suggests follow-up recommendations.
4	Therapist ends session and adds notes.	System securely saves session notes and updates patient records.
Alternate Course of Action		
S#	Actor Action	System Response
1	Internet connection drops during session.	System pauses and attempts auto- reconnect.
2	Therapist disables AI assistance.	System logs only manual notes without AI suggestions.
3	Patient exits session early.	System marks session incomplete and notifies therapist.

Table: UCS 2

Explanation: This use covers the encrypted online therapy process and AI support, ensuring privacy, personalization, and continuity in therapy.

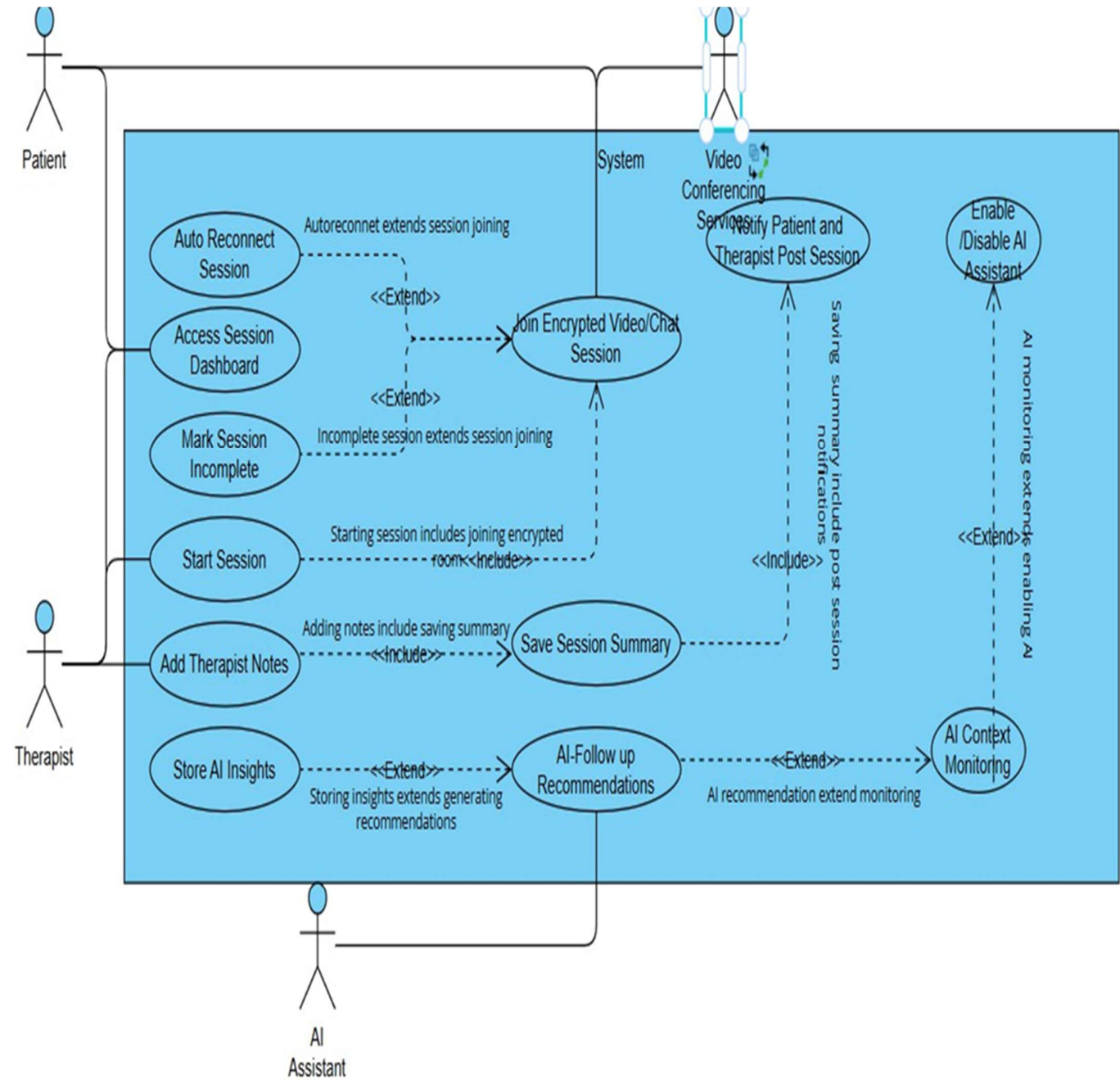


Figure: use case 2

3.3 Manage Patient Records and Notes

Identifier	UC-1	
Purpose	To allow therapists to create, update, and securely store patient medical histories, reports, and session notes.	
Priority	High	
Pre-conditions	Therapist must be logged in and authorized.	
Post-conditions	Updated patient records stored securely in database.	
Typical Course of Action		
S#	Actor Action	System Response
1	Therapist logs in and accesses patient profile.	System displays patient history and reports.
2	Therapist uploads new reports or session notes.	System encrypts and stores files securely.
3	Therapist edits or updates medical information.	System validates and updates data.
4	Therapist saves changes.	System confirms successful update.
Alternate Course of Action		
S#	Actor Action	System Response
1	Therapist uploads unsupported file type.	System displays an error message.
2	Database temporarily unavailable.	System queues update and retry later.

Table: use case 3

Explanation: This use case ensures secure handling of sensitive medical records with proper encryption and validation.

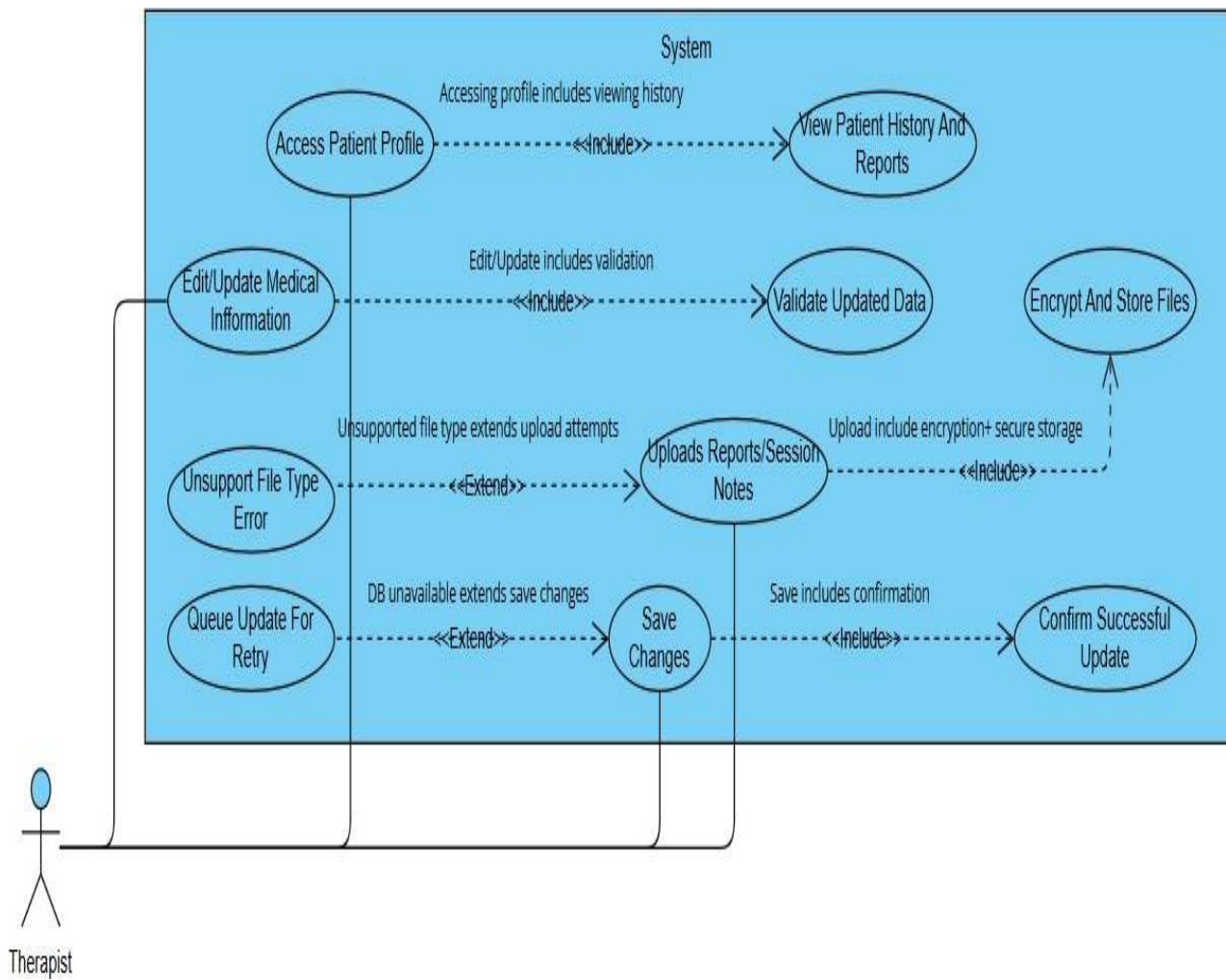


Figure: use case 3

3.4 Assign and Track Wellness Goals:

Identifier	UC-1	
Purpose	To enable therapists to assign wellness or therapeutic goals and allow patients to track progress.	
Priority	Medium	
Pre-conditions	Both patient and therapist must be registered and linked.	
Post-conditions	Progress data recorded and viewable in dashboards.	
Typical Course of Action		
S#	Actor Action	System Response
1	Therapist assigns a wellness goal to the	System store’s goal details and due date.
	patient.	
2	Patient views assigned goals on dashboard.	System displays goal progress indicators.
3	Patient marks goal as completed.	System updates completion percentage and awards points.
4	Therapist reviews progress.	System shows visual analytics and improvement trends.
Alternate Course of Action		
S#	Actor Action	System Response
1	Patient misses deadline.	System sends automated reminder notification.
2	Therapist deletes a goal.	System updates both dashboards and logs change.

Table: use case 4

Explanation: This case promotes engagement and motivation through gamified goal tracking and progress monitoring.

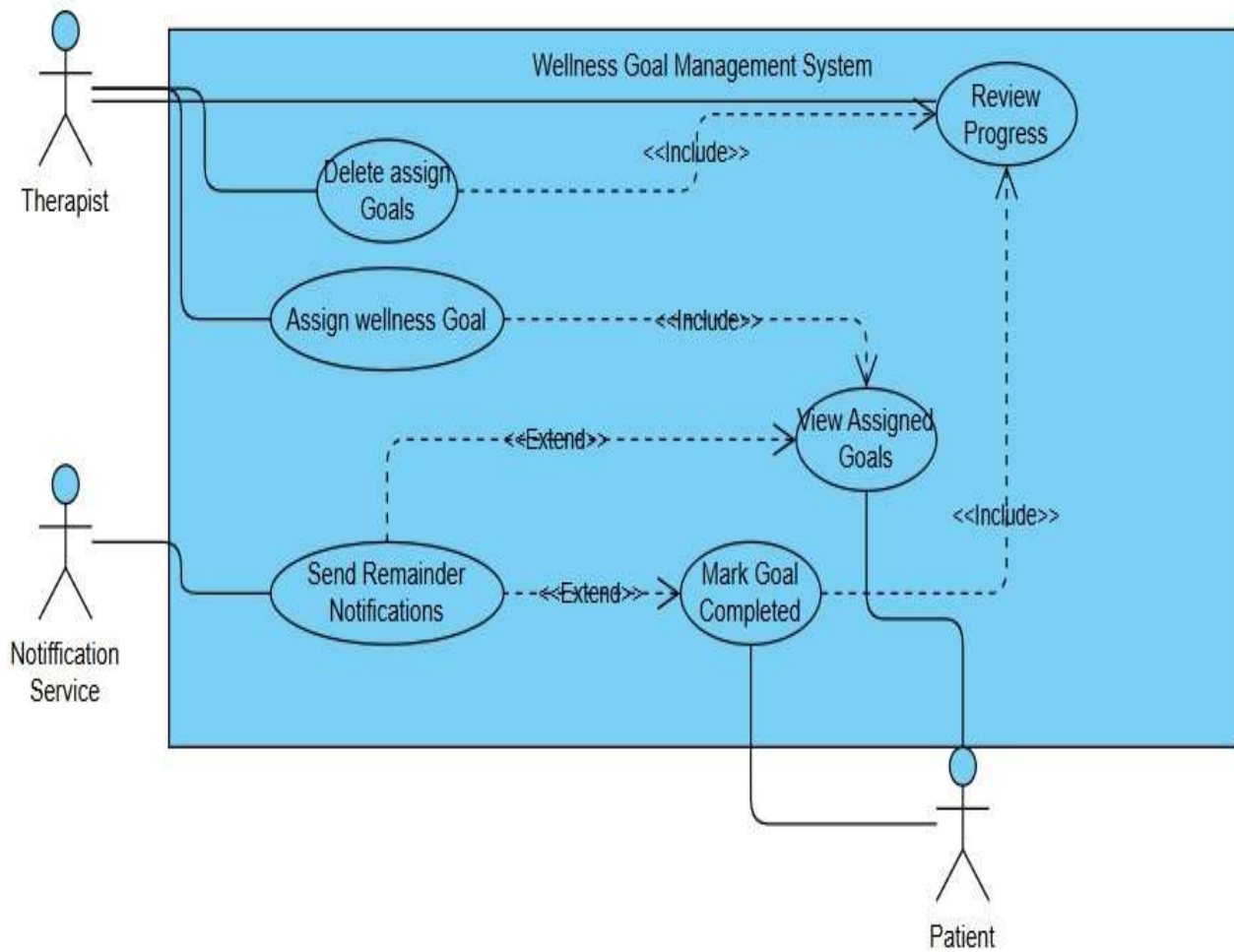


Figure: use case 4

3.5 Interact with AI Virtual Assistant

Identifier	UC-1	
Purpose	To enable patients to communicate with an AI-powered assistant that provides personalized recommendations and reminders based on patient data	
Priority	High	
Pre-conditions	Patient logged in and AI assistant enabled.	
Post-conditions	Personalized recommendations logged and optionally shared with therapist.	
Typical Course of Action		
S#	Actor Action	System Response
1	Patient opens AI chat window.	System activates virtual assistant module.
2	Patients ask for advice or progress summary.	AI analyzes profile data and generates a context-aware response.
3	AI assistant provides tailored suggestions.	System displays personalized recommendations (e.g., exercises, reminders).
4	Patients save useful advice.	System logs interaction for later review.
Alternate Course of Action		
S#	Actor Action	System Response
1	Patient asks an out-of-scope question.	AI responds with polite clarification or resource link.
2	AI module temporarily offline.	System displays "Assistant unavailable, please try later."

Table: use case 5

Explanation: This case integrates AI to deliver smart, real-time support while maintaining privacy and personalization.

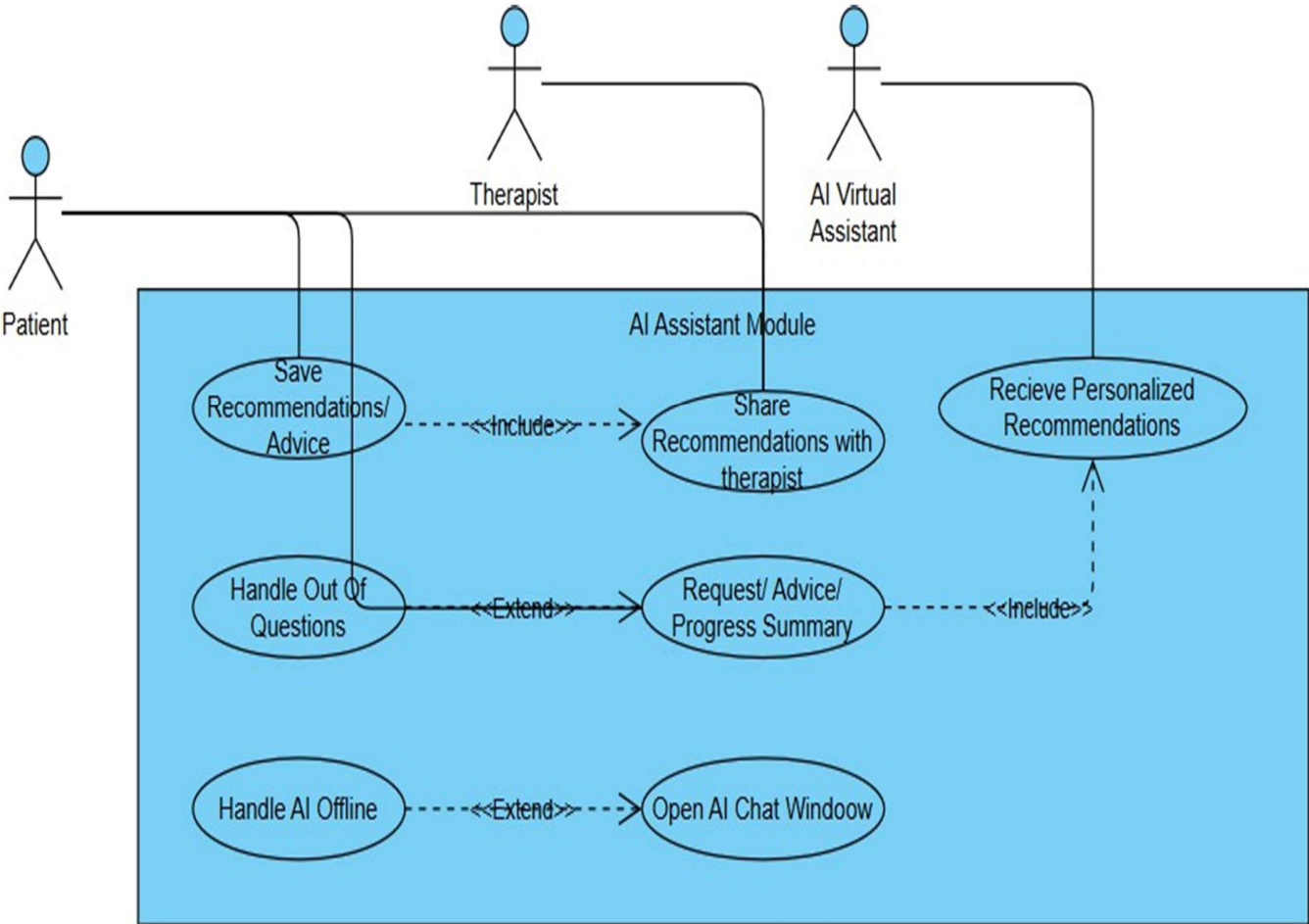


Figure: use case 5

3.6 Manage User Accounts and Subscriptions

Identifier	UC-6	
Purpose	To allow Admin to manage user accounts, therapist verification, and subscription plans.	
Priority	Medium	
Pre-conditions	Admin must be authenticated.	
Post-conditions	Updated subscription and account data stored securely.	
Typical Course of Action		
S#	Actor Action	System Response
1	Admin logs into the Admin Panel.	System displays list of users and subscriptions.
2	Admin verifies a new therapist's credentials.	System marks therapist as 'Verified.'
3	Admin updates or terminates a subscription.	System applies change and notifies affected users.
4	Admin checks platform health reports.	System displays real-time analytics.
Alternate Course of Action		
S#	Actor Action	System Response
1	Admin attempts invalid operation.	System restricts action and shows warning message.
2	Network error during update.	System logs failure and retries automatically.

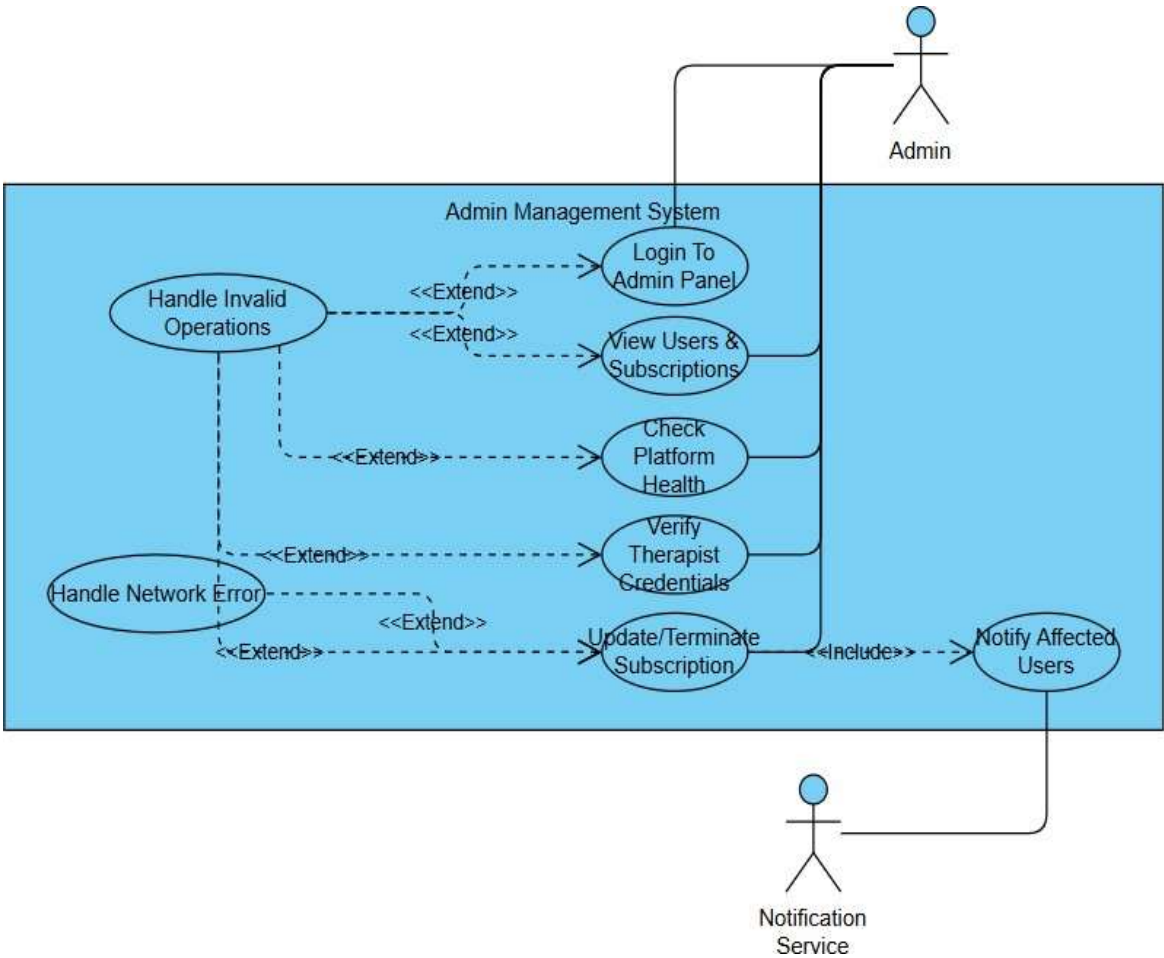
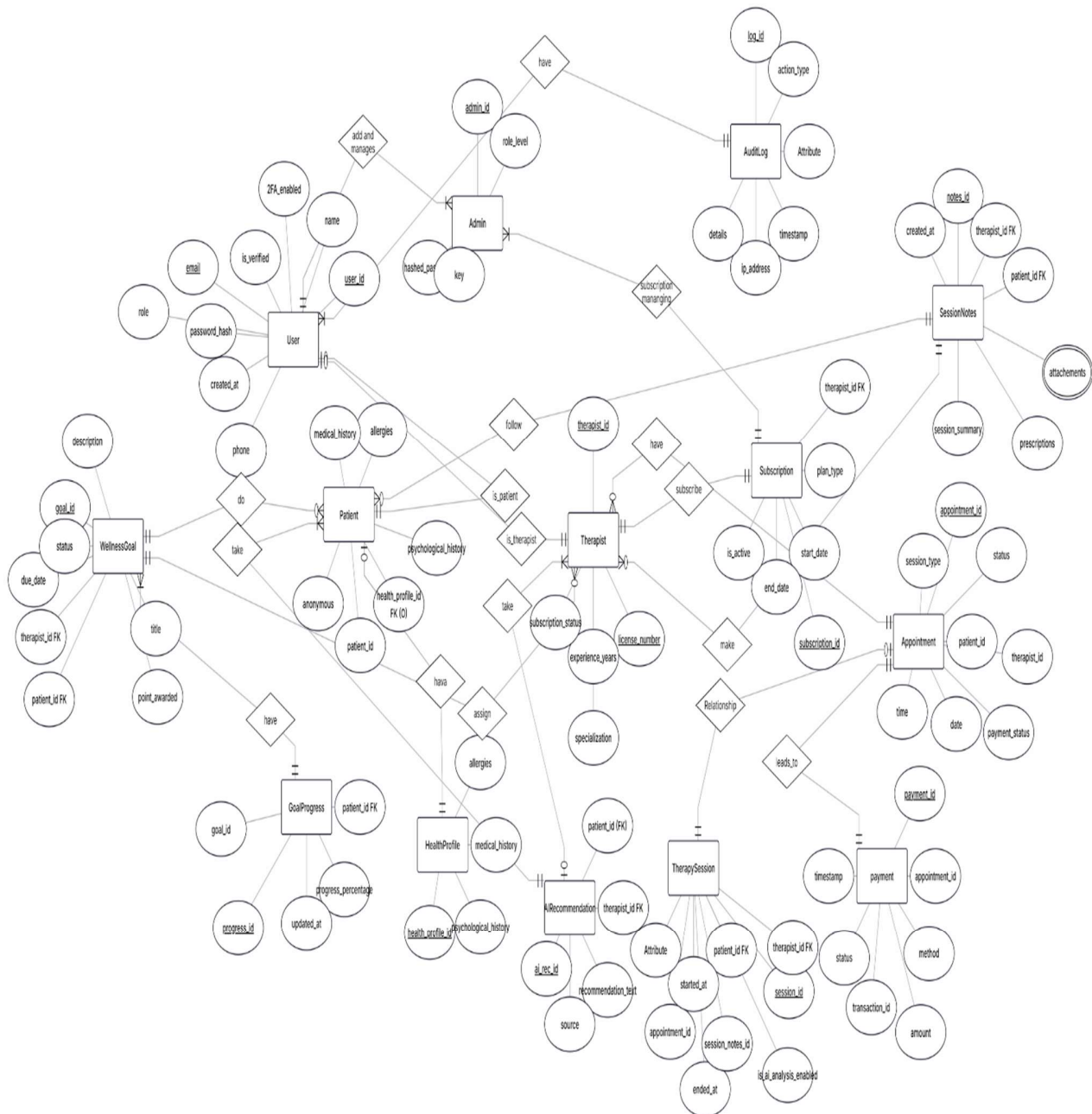


Table: use case 6

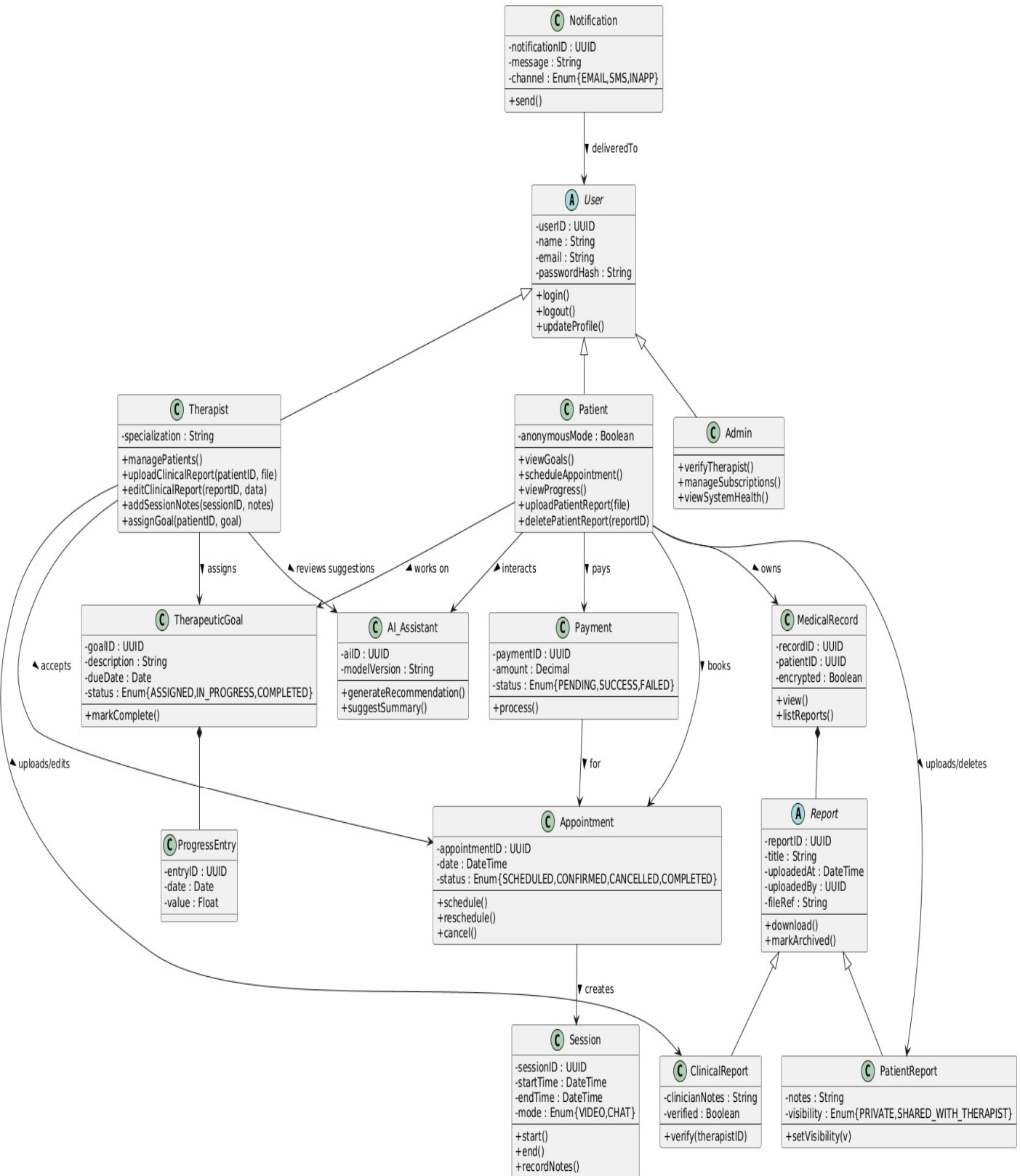
Explanation: This use case maintains the integrity and accessibility of the platform through verified therapist registration and subscription control.

3.7 Requirements Analysis and Modeling

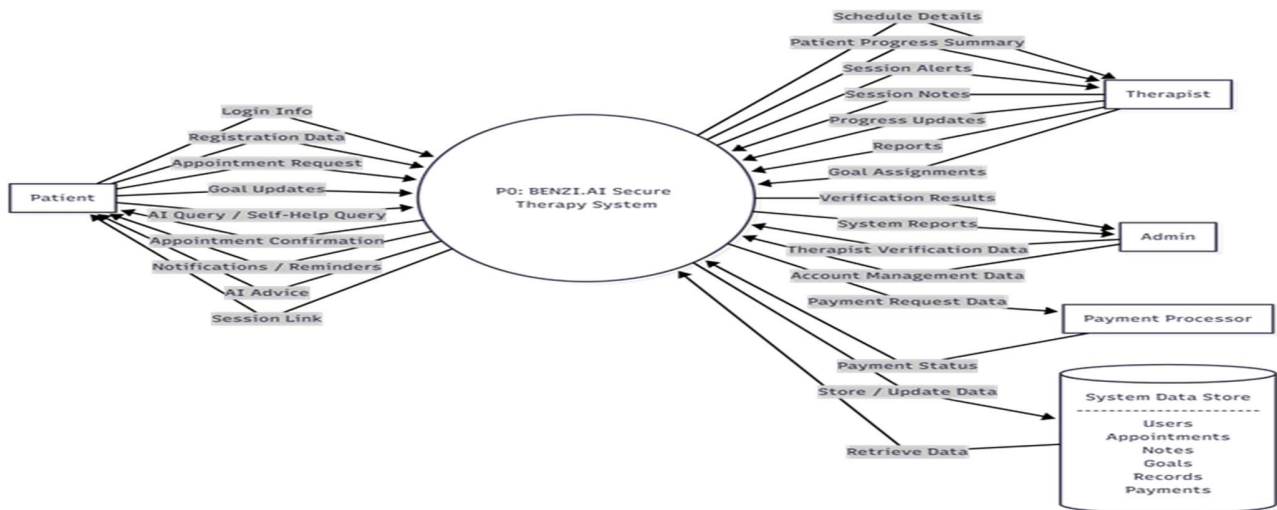
Figure ERD:



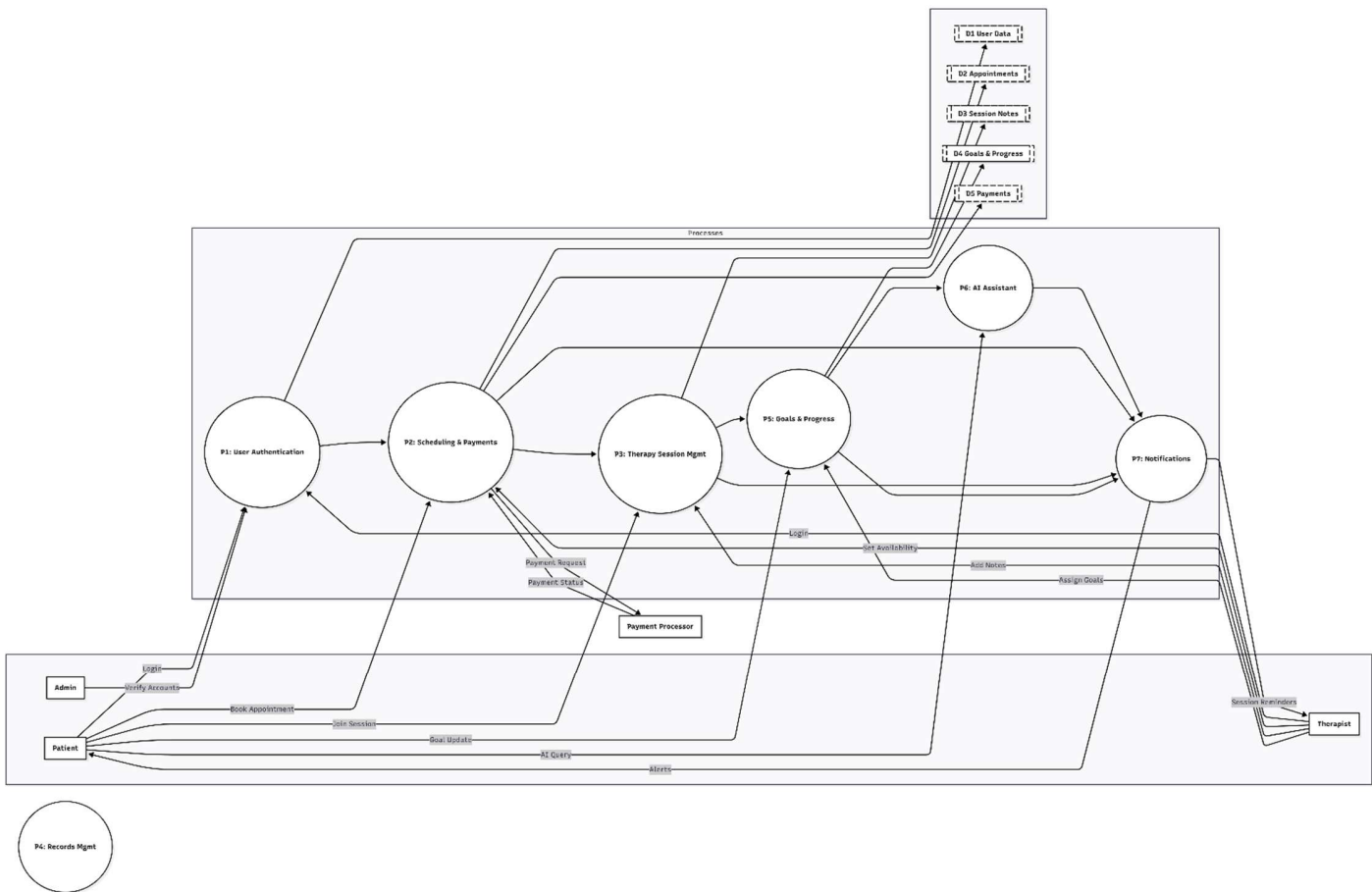
Abstract Class Diagram:



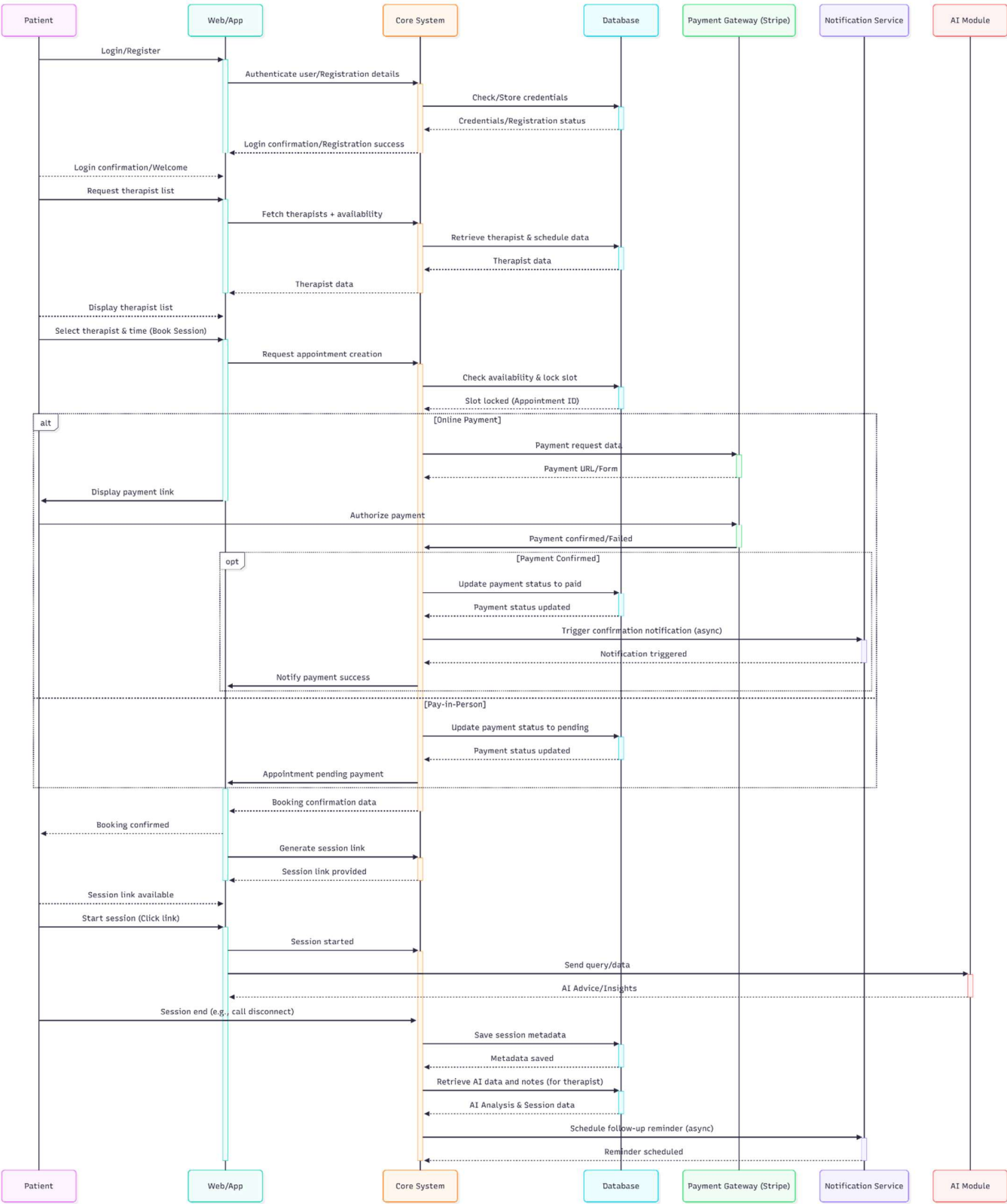
DFD:
Level 0:



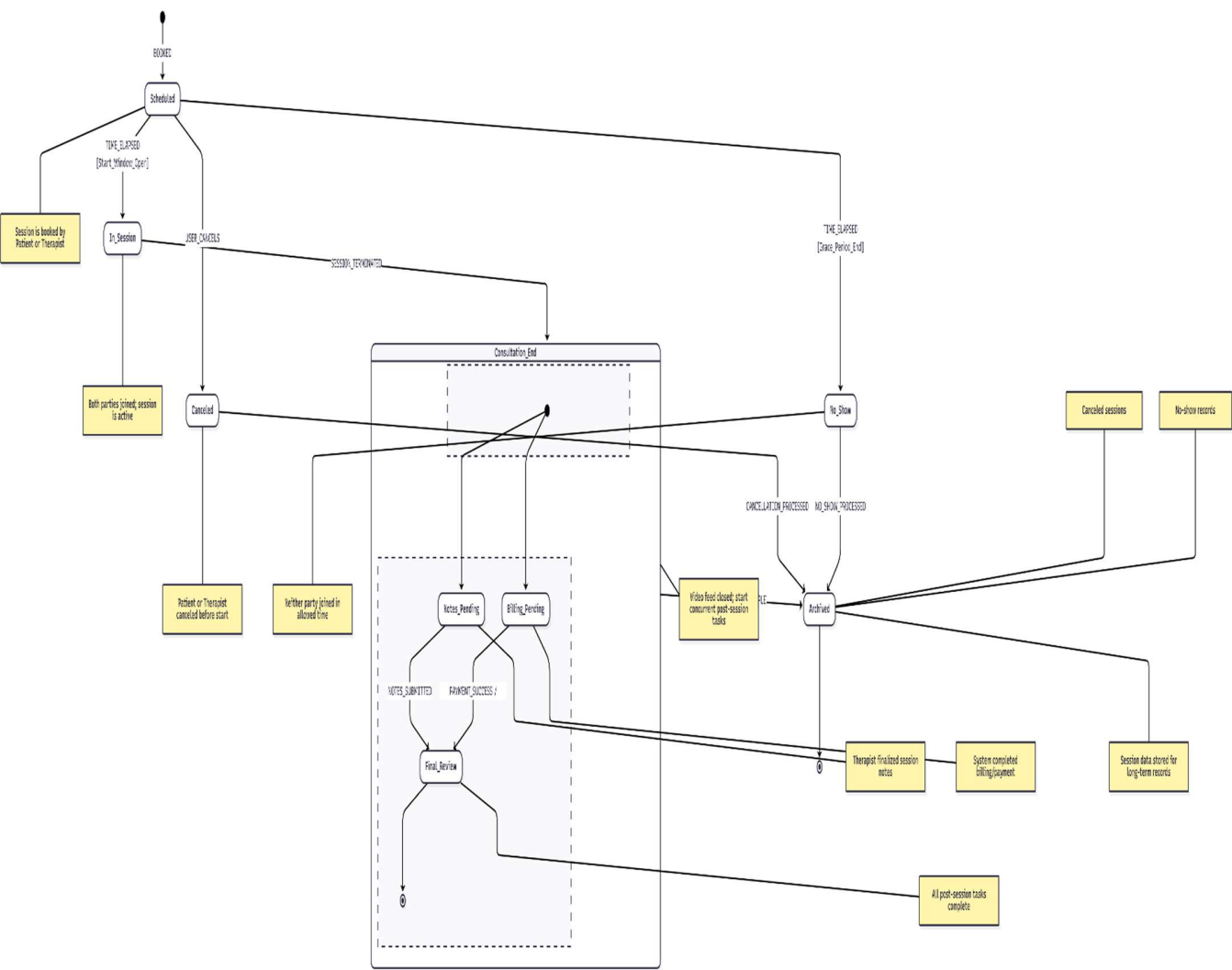
Level 1 :



Sequence Diagram:



State Diagram:



4 Nonfunctional Requirements

4.1 Performance Requirements

Consistent responsiveness across all modules must be guaranteed by the AI-Integrated Secure Therapist–Patient Management System. To ensure a smooth user experience, the system should load the main dashboards, login pages, and appointment interfaces in three seconds under steady network conditions. To guarantee that users don't notice any delays, all interactive tasks, including authentication, appointment scheduling, and feature navigation, must finish in about two seconds. The latency of real-time chat and messaging features must be kept to a minimum so that patients and therapists can communicate with each other without any delays of more than one second. To maintain a smooth conversational flow without interfering with therapy sessions, the AI Virtual Assistant should also reply to user questions or suggestions in less than four seconds.

4.1.1 Concurrent User Handling

Without sacrificing speed or dependability, the platform must be able to accommodate many concurrent users. The system should effectively manage up to 500 concurrent users in all roles administrators, therapists, and patients while preserving consistent performance metrics during normal operation. The system should use cloud-based auto-scale to scale up to 1,000 concurrent users to handle high-demand usage scenarios, like therapy sessions or extensive testing. It's important to keep resource usage within reasonable bounds, making sure that memory usage stays below 75% and CPU usage doesn't go above 70%. During concurrent user interactions, these performance levels ensure consistent responsiveness and avoid performance degradation.

4.1.2 Database Efficiency

Maintaining seamless system operation is largely dependent on database performance. The creation, retrieval, updating, and deletion of patient records, appointments, and therapy logs, among other database operations, should all be completed quickly ideally, in less than a second per query. For collections that are frequently accessed, like appointments, session logs, and AI recommendations, the system must make use of appropriate indexing and query optimization strategies to improve data access speed. The encryption overhead must be kept to a minimum, adding no more than 20% to the query execution latency, because the database uses client-side field-level encryption for sensitive data. To avoid performance snags during periods of high usage, the database architecture should be tuned for parallel read and write operations.

4.1.3 AI Module Processing

To guarantee prompt delivery of AI-driven insights and recommendations, the Python-developed integrated artificial intelligence module must function effectively. Under typical load circumstances, the AI assistant should produce tailored suggestions for each request in two to four seconds. Even with high data volumes, the AI subsystem should finish analysis in ten seconds when processing aggregated data for behavioural trend summaries, goal tracking analytics, or performance reports. Workflows that are automated and in charge of scheduling tasks, sending reminders, and initiating AI-driven events should operate consistently and with a maximum delay of five seconds from their specified trigger time. This guarantees that user actions and automated background processes are synchronised in real time.

4.1.4 Real-Time Communication Performance

For therapist-patient interactions to continue unbroken, the system's communication performance is essential. To maintain visual and aural clarity, all video sessions held using integrated APIs like Zoom or Google Meet must have a minimum resolution of 720p at 30 frames per second. Under typical broadband circumstances, the latency during live video or chat sessions cannot be more than 300 Ms. To reduce session disruption, the system should automatically try to reconnect within five seconds of any brief network outages. Transmission speed shouldn't be negatively impacted by the secure communication protocols used, guaranteeing stable, encrypted, and seamless therapy sessions over time.

4.1.5 System Availability and Throughput

For all users to have consistent access, the system must maintain high availability. With planned maintenance and upgrades excluded, an uptime of at least 99.5 percent is necessary. At least 100 requests per second should be processed by the backend, and average API response times shouldn't be more than 500 Ms. Under a steady internet connection, file uploads, like prescriptions or clinical reports, that are no larger than five megabytes, must finish in about five seconds. Continuous system accessibility and satisfying user experience are guaranteed by the combination of dependable data transmission and optimized server response times.

4.1.6 Scalability and Efficiency

Horizontal scaling should be supported by the cloud infrastructure that houses the platform in order to handle growing user demand and data growth without experiencing any outages. To ensure continuous service delivery during traffic spikes, the system must be built to dynamically allocate additional computational resources. Caching techniques like Redis or in-memory caching should be used to store static data and AI pre-processed responses in order to improve performance efficiency. This method improves the responsiveness of frequently accessed modules, minimises API latency, and cuts down on repetitive computations. Together, these caching and scalability techniques guarantee that the system will continue to be flexible, effective, and able to sustain fast performance even as the number of users rises.

4.1.7 Rationale

In order to ensure a responsive, effective, and reliable system that fits with the delicate and time-sensitive nature of digital therapy, these performance requirements were set. By ensuring low communication latency, quick database access, and optimized AI response times, therapists can conduct sessions with ease, patients receive care without interruption, and administrative procedures run smoothly. Developers can make well-informed design and optimization decisions throughout implementation thanks to the established benchmarks, which offer a quantifiable standard for assessing system performance.

4.2 Safety Requirements

4.2.1 Data Integrity and Loss Prevention

The processing, transmission, and storage of sensitive data must be done without any data loss or corruption, thanks to an AI-Integrated Secure Therapist–Patient Management System. Duplicate database backups and automated recovery procedures must be used to store all patient and therapist data, including medical histories, therapy notes, and session records. To reduce the chance of data loss in the event of a server malfunction or system failure, regular encrypted backups will be planned at predetermined intervals. To guarantee that interrupted or partial updates do not jeopardise the accuracy of stored data, the database system will employ atomic operations to preserve transactional integrity. The system must be able to fully restore from the most recent backup without jeopardising confidentiality or data structure consistency in the case of a hardware or software failure.

4.2.2 System Fault Tolerance and Failure Recovery

The platform must be built with built-in fault tolerance mechanisms to guarantee secure operations and uninterrupted therapy sessions. To avoid complete system failures during component failures, load balancers, redundant servers, and automated failover systems will be set up. Critical functions like secure login, patient data retrieval, and appointment scheduling must be maintained even if a subsystem such as the AI module, video conferencing integration, or automation service becomes unavailable. All failure events must be recorded by automatic

error logging and exception handling to support post-incident diagnostics and avoid future occurrences of the same failures. To avoid corrupted therapy records or incomplete data writes, the system should also offer safe session termination in the event of unplanned crashes.

4.2.3 User and Patient Safety

In terms of data security and mental health, the system must put the safety of therapists and patients first. Safeguards must be put in place to prevent unauthorized disclosure of personal information or medical details because the platform handles sensitive mental health data. Patients interacting with the AI assistant or participating in therapy sessions need to be shielded from inadvertently meeting upsetting or dangerous material. The AI module will function within predetermined ethical guidelines, guaranteeing that its suggestions stay non-diagnostic, supportive, and consistent with accepted therapeutic approaches. When the AI recognizes phrases that might indicate distress or self-harm, it should set off an automated alert system that securely connects the user to emergency services or notifies the therapist without saving private information in plain text.

4.2.4 Secure Communication and Session Safety

Complete end-to-end encryption must be maintained for all real-time communication, including voice, video, and text messages, to guard against data misuse or interception. The system must use secure session verification procedures and encrypted tokens to prevent eavesdropping, screen recording, and unwanted session access. Users must not be exposed to personally identifiable information or system flaws that could jeopardize their privacy when interacting in anonymous mode. The application will make sure that all modifications to therapy notes, or medical records only take place through validated and authenticated workflows, preventing risky operations like data modification during live video sessions.

4.2.5 Hardware, Network, and Operational Safety

International safety and reliability standards must be followed when deploying the hosting infrastructure in monitored, secure data centres. Every piece of hardware and network should be shielded from environmental dangers, power surges, and unwanted physical access. The cloud provider is required to guarantee adherence to established operational safety standards, including SOC 2 for operational controls and ISO/IEC 27001 for information security management. To stop malicious data injection attempts or denial-of-service attacks, network firewalls, intrusion detection systems, and rate-limiting mechanisms must be installed. To prevent system overload or unplanned downtime, the system should also monitor CPU temperature, memory usage, and network bandwidth in order to enforce safe operating thresholds.

4.2.6 Regulatory and Ethical Compliance

The platform must abide by the legal and ethical guidelines that control how medical and psychological data is handled. International data protection standards, such as those found in the General Data Protection Regulation (GDPR) and the Health Insurance Portability and Accountability Act (HIPAA), must be incorporated into its design. These compliance guidelines guarantee that users have the right to view, update, or remove their personal data and that patient information is gathered, saved, and processed with their express consent. To ensure ongoing adherence to these standards, regular vulnerability assessments and safety audits must be carried out. To avoid human error that could result in safety incidents, all staff members engaged in development or maintenance should receive the proper cybersecurity and data handling training.

4.2.7 Rationale

Beyond technical dependability, safety in a mental health management system includes user protection, ethical data handling, and failsafe recovery in the event of an emergency. The safeguards are intended to maintain therapy services while preventing harm to the body, mind, and information. With rigorous adherence to encryption protocols, backup plans, and compliance frameworks, the system ensures a safe and regulated environment that upholds patients' and therapists' confidence.

4.3 Security Requirements

4.3.1 Data Protection and Encryption

Through extensive encryption and access control measures, the AI-Integrated Secure Therapist–Patient Management System must guarantee the highest degree of data availability, confidentiality, and integrity. Transport Layer Security (TLS 1.3) and secure cypher suites must be used to protect all data in transit between clients and servers in order to guard against interception or tampering during communication. Field-level encryption based on AES-256 [3] [16] standards must be used to protect sensitive data stored in the database, such as patient records, therapy notes, prescriptions, and session logs. For sensitive data, client-side encryption must be used to prevent even system administrators from accessing plaintext data. To prevent exposure during storage or recovery procedures, backups and archived data must also be encrypted while at rest.

4.3.2 Authentication and Access Control

To avoid unwanted access, user identity management must adhere to robust authentication procedures. Administrators, therapists, and patients must all authenticate using a secure login system that uses hashed credentials using algorithms like Argon2 or Bcrypt. To separate permissions based on user roles, Role-Based Access Control (RBAC) will be implemented across the platform. Therapists can view information about their assigned patients, administrators can manage user accounts without seeing any sensitive health records, and patients can only access their personal data. Therapist and administrator accounts must have Two-Factor Authentication (2FA) enabled for added security. This can be done via SMS verification or Time-based One-Time Passwords (TOTP). Short-lived, safely stored in HTTP-only cookies, and automatically invalidated upon logout or inactivity are requirements for session tokens used for authentication.

4.3.3 Authorization and Session Management

All modules of the system must guarantee safe and traceable user sessions. To prevent token forgery, each session will be identified by a digitally signed JSON Web Token (JWT) that is validated on each request and contains encrypted claims. To reduce the possibility of unwanted access on shared or public devices, sessions will automatically end after a predetermined amount of inactivity. In order to prevent possible account misuse, concurrent session restrictions must be implemented to prevent multiple active logins from the same user credentials in various locations. Both the client and the server must have complete session termination guaranteed by logout mechanisms. For security auditing purposes, session logs should record IP addresses, device fingerprints, and login times without disclosing private information.

4.3.4 Secure Communication and API Protection

To avoid man-in-the-middle attacks, all system communications, both internal and external, must be secured via HTTPS endpoints with certificate pinning. Token-based authentication and IP whitelisting for administrative endpoints will be used to safeguard the backend API. To stop brute-force or denial-of-service attacks, rate limiting and request throttling techniques must be used. To avoid injection attacks like SQL or NoSQL injections, cross-site scripting (XSS), and cross-site request forgery (CSRF), APIs must sanitise all inputs. To lessen browser-based threats, security headers such as Content-Security-Policy (CSP), Strict-Transport-Security (HSTS), and X-Frame-Options must be set up. Strict authorisation checks must be performed on each API call involving patient or therapist data to guarantee adherence to the least privilege principle.

4.3.5 Audit Trails and Monitoring

To keep track of all platform security-related activity, a continuous monitoring and audit logging system must be put in place. Successful and unsuccessful login attempts, data changes, permission adjustments, and administrative actions must all be documented. For future verification, audit records must be unchangeable and kept in a tamper-proof, encrypted repository. To identify irregularities, send out alerts, and start automated incident responses when suspicious activity is found, the system ought to be integrated with a Security Information and Event Management (SIEM) program. Rapid detection and mitigation of possible security breaches or misuse will be made possible by these monitoring capabilities. To maintain a proactive security posture, the continuous integration and deployment (CI/CD) pipeline needs to include automated security scans, vulnerability assessments, and dependency checks on a regular basis.

4.3.6 Data Privacy and Regulatory Compliance

International standards and laws protecting the privacy of healthcare data must be adhered to by the platform's security architecture. The system must adhere to the data protection guidelines set forth in the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR). Data minimization, clear user consent for data collection, safe storage, and the right to request data deletion are some of these principles. For AI model training and analytics, the system must guarantee that data is anonymized and pseudonymized to avoid exposing personally identifiable information (PII). In compliance with moral data management principles, users must have complete control over their data, including the option to amend or remove their profiles.

4.3.7 Infrastructure and Physical Security

Secure cloud environments like AWS, Azure, or Google Cloud, which offer built-in compliance with ISO/IEC 27001 and SOC 2 Type II security certifications, are required to host the application and database servers. Virtual private networks (VPNs) and SSH key-based authentication should be used to limit access to server instances. To protect against network-level threats and unauthorized access, firewalls, intrusion detection systems (IDS), and intrusion prevention systems (IPS) must be installed. All administrative actions must require multi-factor authentication, and system administrators must follow the least privilege principle. Certified security experts must perform periodic penetration testing to find and fix vulnerabilities

in the infrastructure and application layers.

4.3.8 Rationale

Strong security measures are essential because the system handles extremely sensitive medical and psychological data. The platform guarantees the confidentiality and protection of all therapist-patient interactions by putting in place multi-layered encryption, stringent access control, and adherence to international security standards. The defence against internal or external threats is further strengthened by ongoing monitoring and auditing, and compliance with HIPAA and GDPR ensures that all data handling procedures continue to be morally and legally sound. Together, these security features guarantee that BENZI.AI [1] upholds user confidence, data integrity, and a safe setting for online mental health services.

4.4 Additional Software Quality Attributes

4.4.1 Usability

User-friendliness and intuitive interaction for all user roles must be given top priority in the AI-Integrated Secure Therapist–Patient Management System. Both technical and non- technical users, including those with low levels of digital literacy, should be able to utilise the interface's intuitive, easily navigable layout. All portals Patient, Therapist, and Admin must have the same navigation structure so that users can easily find essential features like appointments, records, and analytics. To allow for users with physical or visual impairments, the system's design must adhere to Web Content Accessibility Guidelines (WCAG) 2.1 standards. The AI assistant and dashboard interfaces should use simple language, icons, and visual indicators to promote engagement and minimize cognitive load. Usability testing should verify that a novice user can finish basic tasks, like scheduling a session or going over therapy notes, in two minutes without help from a third party.

4.4.2 Reliability

Keeping users' trust is essential, especially in settings that provide mental health support. The system must function consistently and steadily in both low- and high-demand scenarios. During system outages, all crucial functions, including appointment scheduling, encrypted communication, and authentication, must run with a minimum 99% success rate. Performance metrics will be continuously tracked by automated monitoring tools, which will also send out alerts in the event of service outages or degradation. To guarantee continuous operation even in the event of unplanned hardware or network failures, recovery measures like redundant servers and failover systems must be put in place. To avoid partial updates or data loss during concurrent activities, transactional integrity should be ensured through atomic operations and data consistency must always be maintained.

4.4.3 Availability

Except for scheduled maintenance windows, the system must always be accessible to users, with an expected upbring of 99.5%. The hosting infrastructure must have load balancing, distributed databases, and redundant servers to sustain this degree of availability and guarantee continuous access. To reduce disruption, users must be informed of planned maintenance and updates at least 24 hours in advance. If a system component fails, automated failover mechanisms will make sure that another instance takes over right away, protecting user sessions and data integrity. For therapists and patients who rely on prompt access to appointments, records, and continuous communication, high availability is essential to preserving trust.

4.4.4 Maintainability

For the platform to manage growing user loads, higher data volumes, and increased system functionality without requiring significant reconfiguration, it must support both horizontal and vertical scalability. With only minor adjustments to current components, the architecture should enable the addition of new modules, APIs, or AI models. For example, it should be possible to integrate future features like wearable device data synchronization, mobile applications, or sophisticated analytics dashboards without having to completely redesign the core system. Additionally, with automated migration scripts, the system should support multiple environments, including development, staging, and production, while maintaining deployment flexibility. These characteristics guarantee long-term flexibility and increase the system's utility as organisational and technological requirements change.

4.4.6 Interoperability

The system must be built to function well in a variety of environments and platforms without requiring a lot of reconfigurations. The web application should be responsive on a variety of devices, including desktops, tablets, and smartphones, and fully compatible with the main browsers, such as Chrome, Firefox, Edge, and Safari. Migration between cloud providers (such as AWS, Azure, or Google Cloud) must be possible with deployment scripts that require little manual intervention. Regardless of underlying hardware or operating system variations, containerization technologies such as Docker will be used to guarantee that the application can run consistently across multiple environments. The software product's long-term sustainability, scalability, and deployment are made simpler by its portability.

4.4.7 Testability

Unit, integration, system, and user acceptance testing are just a few of the many levels of comprehensive and repeatable testing that the system must be built to support. Frontend functionalities will be tested using Cypress or Selenium, while backend logic will be validated using automated testing frameworks like Jest and Mocha. Critical scenarios like data encryption, video session stability, AI response accuracy, and authentication must all be covered in test cases. To enable independent testing, each software module should have precise definitions for input and output. Regression tests should be automatically run by Continuous Integration (CI) pipelines following each code update to make sure that new features don't cause flaws in already existing functionalities. Maintaining a high standard of software correctness and reliability requires regular review and logging of test results.

4.4.8 Robustness

By including these software quality attributes, the system is guaranteed to be dependable, flexible, and user-focused for the duration of its lifecycle. While maintainability and scalability ensure long-term sustainability and technological relevance, usability and reliability have a direct impact on patient trust and adoption. For safe digital healthcare settings, the AI-Integrated Secure Therapist–Patient Management System strikes a balance between performance, adaptability, and resilience by following industry standards and contemporary software engineering principles.

5 Other Requirements

5.1 Database Requirements

- MongoDB 6 or later will serve as the system's main database.
- Information will be kept in several collections: AI Recommendations, Therapy Notes, Goals, Payments, Users, and Appointments.
- Using AES-256 keys, client-side field-level encryption (FLE) will protect all private information, including the patient's name, diagnosis, and session notes.
- Every day encrypted backups will be planned and stored for 30 days; every month, automated restoration tests will be carried out.
- To guarantee query times < 1s, indexing will be applied to frequently queried fields (such as email and appointment date).

5.2 External Interface Requirements

5.2.1 Hardware Interfaces

- Servers hosted in the cloud that have at least four virtual CPUs and eight gigabytes of RAM per instance.
- End-user devices: desktops, laptops, or smartphones that support encrypted video sessions and have a webcam and microphone.

5.2.2 Software Interfaces

- **Zoom SDK/Google Meet API:** for sessions of therapy that are encrypted.
- For safe online payments, use the Stripe [12] Sandbox API.
- Email/SMS gateways for reminders and notifications (SendGrid or Twilio [13]).
- **Automation Server:** used to orchestrate workflows and schedule AI tasks.
- KMS (AWS KMS or Azure Key Vault) [7]: for safe key rotation and storage. Interfaces for communication

- With HSTS enabled, HTTPS (TLS 1.3) will be used for all network communications.
- WebSocket connections will be encrypted using the WSS protocol, and JWT tokens will secure API calls.

5.3 Requirements for Localisation and Internationalisation:

- The platform's user interface and AI responses will initially support English.
- Localization files for Arabic and Urdu, using UTF-8 encoding, will be included in future releases.
- Using browser settings, date/time formats will adjust to the user's location (MM/DD/YYYY vs. DD/MM/YYYY).
- To make future expansion easier, all static text will be externalized into translation files.

5.4 Legal and Regulatory Requirements

- The system design will ensure that personal and health data is processed lawfully by adhering to global privacy principles comparable to HIPAA and GDPR.
- Prior to creating an account, users will expressly consent to data collection.
- Patients will be able to request that their personal data be deleted through a "Right to Erase" function.
- Logging and audit trails will be unchangeable to support legal investigations if necessary; third-party APIs will be examined for adherence to ethical AI standards and data protection laws.

5.5 Ethical and AI Governance Requirements

The AI assistant only offers wellness recommendations and reminders; it will not provide diagnoses or treatments.

- To stop harmful, biased, or medically incorrect content from spreading, AI outputs will be filtered.
- To eliminate personally identifiable information (PII), all datasets used to train AI models will be pseudonymized.
- Reports on AI fairness, bias, and transparency will be examined in annual audits.

5.6 System Reuse and Extensibility Objectives

- The platform's modular architecture (Node/Express backend + React frontend + n8n automation) allows for the future reuse of parts like
- Modules for payment and appointment scheduling in other areas of healthcare.
- A stand-alone chatbot microservice with AI support.
- To facilitate seamless integration with third-party wellness or hospital management systems, APIs will adhere to REST + OpenAPI 3.0 standards.
- For internal academic reuse, the codebase will be kept up to date under an MIT-compatible license.

6 Revised Project Plan

6.1 Progress Overview

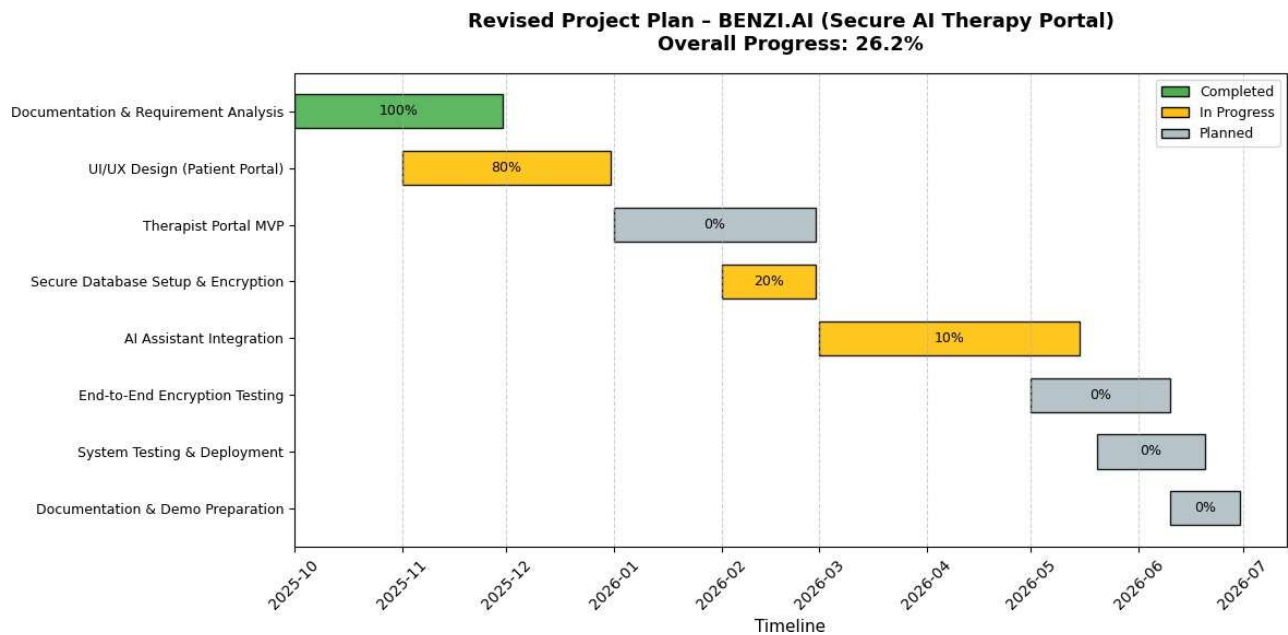
With minor modifications to the development sequencing, the project "BENZLAI – Secure AI Therapy Portal" is moving forward as planned. The Patient Portal UI/UX is almost complete, and all documentation and requirement analysis tasks have been finished ahead of schedule. The creation and verification of the database schema design represent a first step towards the deployment of encryption. The AI Assistant Module's initial design and research have started.

As of November 2025, the project's overall progress is projected to be about 30% of the entire system development effort.

6.2 Phase-Wise Progress Summary

Phase / Task	Planned Duration (Proposal)	Actual / Revised Duration	Progress (%)	Remarks / Status
Documentation & Requirement Analysis	Oct 2025 – Feb 2026	Completed (Nov 2025)	100 %	SRS, Use-cases, and Diagrams finalized
UI/UX Design (Patient Portal)	Nov – Dec 2025	In Progress	80 %	Dashboard prototype ready
Therapist Portal MVP	Jan – Feb 2026	Planned	0 %	To start Dec end
Secure Database Setup & Encryption	Feb 2026	In Progress	20 %	Schemas Creation Done
AI Assistant Integration	Mar – May 2026	Planned	10 %	Started working on Model Architecture
End-to-End Encryption Testing	May – Jun 2026	Planned	0 %	To be done after AI integration
System Testing & Deployment	May – Jun 2026	Planned	0 %	Pending
Documentation & Demo Preparation	Jun 2026	Planned	0 %	Pending

6.3 Gant Chart:



6.4 Updated Plan Highlights

- SRS documentation and requirement analysis were finished ahead of schedule.
- **UI/UX Progress:** The dashboard is 80% operational and prepared for integration.
- **Database Phase:** Encryption will be implemented after the schema is finished.
- **AI Assistant:** Model architecture and research were started.
- **Upcoming milestones (December 2025–February 2026):** Complete the user interface for the patient portal, start the therapist portal, and encrypt the database.
- The ultimate goal is to test and deploy the system by June 2026, along with a project demo and presentation.

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Appendix A: Glossary

Term	Definition / Description
BENZI.AI	<i>Proposed Secure AI Therapy Portal that enables encrypted communication between patients and therapists and integrates an AI wellness assistant.</i>
SRS	<i>Software Requirements Specification – document describing system functionality, constraints, and interfaces.</i>
MVP	<i>Minimum Viable Product – initial version with core features for validation.</i>
Patient	<i>The end-user receiving therapy or wellness guidance via BENZI.AI.</i>
Therapist	<i>Licensed professional using the portal to manage sessions, notes, and communication with patients.</i>
Admin	<i>System administrator responsible for managing accounts, monitoring performance, and maintaining data security.</i>
UI/UX	<i>User Interface / User Experience – design considerations for usability and accessibility.</i>
AI Assistant	<i>Machine-learning module that offers therapy reminders and self-care suggestions; not diagnostic.</i>
MongoDB	<i>NoSQL database used to store and manage encrypted data collections.</i>
Field-Level Encryption (FLE)	<i>Technique used to encrypt sensitive fields directly in the application before storing them in the database. [5]</i>
KMS (Key Management Service)	<i>Secure platform (e.g., AWS KMS) for managing encryption keys. [6]</i>
AES-256	<i>Advanced Encryption Standard with 256-bit keys used for data encryption. [2]</i>
JWT (JSON Web Token)	<i>Token format for secure authentication between client and server.</i>

2FA Factor Authentication) (Two-	<i>Additional security measure requiring a second verification step.</i>
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Term	Definition / Description
TLS / HTTPS	Encryption protocol ensuring secure data transmission over the internet.
HIPAA	U.S. Health Insurance Portability and Accountability Act that defines healthcare data privacy and security standards. [3]
GDPR	EU General Data Protection Regulation outlining user rights and privacy obligations. [1]
OWASP ASVS	Application Security Verification Standard – security checklist for testing web applications. [4]
RBAC	Role-Based Access Control – defines permissions based on user roles (Admin, Therapist, Patient).
E2EE (End-to-End Encryption)	Encryption model ensuring only sender and receiver can read communication data. [7]
WebRTC	Protocol for real-time audio/video communication over browsers. [8]
Zoom SDK / Google Meet API	Tools used to embed secure video conferencing capabilities into BENZI.AI. [7]
n8n	Workflow automation platforms are used for scheduling and process orchestration. [9]
OpenAPI	Specification format (Swagger) for documenting RESTful APIs.
REST API	Interface using HTTP requests data access and manipulation between frontend and backend.
CI/CD	Continuous Integration and Continuous Deployment – pipeline for automated testing and updates.
Docker	Containerization technology for consistent deployment environments.
Audit Trail	Secure record of user activities for accountability and compliance.
Anonymization / Pseudonymization	Process of removing or substituting identifiers to protect personal data privacy.

Term	Definition / Description
Right to Erasure	GDPR right allowing users to request deletion of personal data. [1]
Data Minimization	Collecting only necessary user data as per GDPR principles. [1]
Consent Management	Mechanism ensuring users' consent before collecting or processing data. [1, 3]
System Testing	Phase verifying system behavior against requirements.
Penetration Testing	Simulated cyberattack testing for vulnerabilities before release. [4]
AES Key Wrap	Standard method for encrypting keys with another key (used in secure key exchange). [2]
FHIR / HL7	Healthcare data exchange standards used for interoperability.
SIEM	Security Information and Event Management – system that monitors and correlates logs for threats.
Privacy by Design	Engineering principle embedding data protection from the start. [1, 3]
Fallback Mode	Feature that allows limited operation if AI or video modules fail.
Version Control (Git)	Tool to track changes and manage collaborative development.
Session Timeout	Automatic logout after inactivity for security reasons.
Scalability	System's ability to handle increased users or load efficiently.

Appendix B: IV & V Report

(Independent verification & validation)
IV & V Resource

Name

Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
...					

Table 2: List of non-trivial defects

